



NumClass User Manual

Installation, Usage, and Reference Guide

Manual Version 1.2

For NumClass version 2.2

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```
Number statistics:
Input:          2**5+10
Number:         42
Digits:         Count=2, Sum=6, Product=8
Parity:         Even
Prime:          No, nearest primes: <41 >43, π(n)=13, density=30.95%
Digital root of |n|: 6 (sequence: 42 → 6, additive persistence: 1)
Mult. persistence: 8 (sequence: 42 → 8)
Zeckendorf:     42 = 34 + 8 (F9 + F6), bits(F9..F2)=10010000
Prime factorization: 2 × 3 × 7
Prime factor counts: Ω(n) = 3, ω(n) = 3
Euler's totient φ(n): 12 φ(2)=1, φ(3)=2, φ(7)=6 ⇒ 1*2*6=12
Carmichael λ(n): 6 λ(2)=1, λ(3)=2, λ(7)=6 ⇒ lcm(1, 2, 6)=6
1/n (base 10): 1/n = 0.0238095 repeats, period 6, preperiod 1
Möbius μ(n): -1 (squarefree, odd number of prime factors)
Radical rad(n): 42 (product of distinct primes)

Divisor statistics of |n|:
Divisor count τ(n): 8
Divisors:       1, 2, 3, 6, 7, 14, 21, 42
Classical div. sums: σ(n)=96, s(n)=σ-n=54, q(n)=s-1=53
Unitary div. sums: σ*(n)=96, s*(n)=σ*-n=54, r*(n)=8, q*(n)=s*-1=53
Aliquot sequence: 42 → 54 → 66 → 78 → 90 → 144 → 259 → 45 → 33 → 15 → 9 → 4 → 3 → 1 → 0 (steps: 14 - peak 259 @ [7] - 1 ms)
Quasi-aliquot seq.: 42 → 53 → 0 (steps: 2 - peak 53 @ [2] - 0 ms)

Arithmetic and Divisor-based
- Abundant number: Proper divisors sum > the given number.
  Details: Sum of proper divisors: 54 > 42.
- Cube-free number: No prime cube divides n (all prime exponents ≤ 2).
  Details: n = 2 × 3 × 7; max exponent = 1 < 3
- Highly abundant number: An integer whose sum of divisors is greater than that of any smaller positive integer.
  Details: σ(42) = 96. Previous record: 36 with σ(36) = 91.
- Practical number: All smaller positive integers can be written as sums of distinct divisors of the number.
  Details: 2 ≤ 1 + σ(2) = 4, 3 ≤ 1 + σ(6) = 13. All conditions satisfied, 42 is practical.
- Primary pseudoperfect number: Satisfies Σ_{p|n} 1/p + 1/n = 1 (sum over distinct prime divisors).
  Details: primes(n) = 2×3×7; 1/2 + 1/3 + 1/7 + 1/42 = 1 (and Σ n/p + 1 = 42 = n)
- Semiperfect number: A positive integer equal to the sum of a subset of its proper divisors.
  Details: 42 = 7 + 14 + 21
- Sphenic number: A positive integer that is the product of exactly three distinct primes.
  Details: 42 = 2 × 3 × 7
- Squarefree number: No prime factor appears more than once in its factorization.
  Details: n = 2 × 3 × 7; max exponent = 1 < 2
- Zumkeller number: Divisors can be partitioned into two equal-sum sets.
  Details: Practical & even σ(n) ⇒ Zumkeller; σ(n)=96, target=48; partition found (MITM); witness: [1, 2, 3, 7, 14, 21] (sum 48)

Combinatorial and Geometric
- Cake number: Max pieces from k planar cuts in 3D: C(k) = (k³ + 5k + 6)//6.
  Details: 42 is the cake number for k=6 straight cuts.
- Catalan number: C_n = binomial(2n,n)/(n+1), counts Dyck paths.
  Details: 42 is Catalan number C(5).
- Lah number: Equals L(n,k) for some n≥1,k≥1: partitions of n elements into k nonempty ordered lists.
  Details: 42 = L(7,6).
- Partition number: Checks if n equals p(k): number of integer partitions of k.
  Details: 42 = p(10).

Digit-based
- Brazilian number: Representable as a repdigit in some base b with 2 ≤ b ≤ n-2.
  Details: In base 4, n is a repdigit of length 3 with digit 2: 2×R_3(4)=(4³-1)/(4-1)
- Harshad number: Base-10 Harshad (Niven) number: divisible by the sum of its decimal digits.
  Details: 42 is divisible by the sum of its digits: 42 ÷ 6 = 7.
- Moran number: Harshad (base 10) with n/s = prime, where s is the sum of decimal digits of n.
  Details: Harshad with s(n)=6; n/s = 7 is prime.
- Odious number: Has an odd number of 1's in binary representation for |n|.
  Details: Binary: 101010, number of 1's: 3 (odd), total bits: 6
- Self number: Cannot be written as m + sum of digits of m for any m.
  Details: 42 cannot be written as m + sum(digits of m) for any m < 42.

Diophantine Representations
- Sum of 3 cubes: Can be written as n = a³ + b³ + c³ for integers a, b, c.
  Details: 42 = (-80538738812075974)³ + 12602123297335631³ + 80435758145817515³
- Sum of 3 squares: Can be written as n = a² + b² + c² for integers a, b, c.
  Details: found 1: 1²+4²+5²

Fun numbers
- Fun number: Numbers that are famous or iconic in pop culture, computing, sci-fi, internet humor, or memes.
  Details: The Answer to the Ultimate Question of Life, the Universe, and Everything (Douglas Adams, The Hitchhiker's Guide to the Galaxy).

Mathematical Curiosities
- Eban number: No letter 'e' in the English spelling.
  Details: 42: forty-two is an eban number (no 'e' in its English spelling)
- Octal-interpretable number: Number can be interpreted as an octal numeral.
  Details: As octal: 42₈ = 34₁₀ = 100010₂ = 22₁₆
- Polydivisible number: In base 10: for every k = 1..d, the first k digits form a number divisible by k.
  Details: Let P_k be the first k digits. For k=1..2, P_k ≡ 0 (mod k). P₁=4 divisible by 1 ; P₂=42 divisible by 2.

Named Sequences
- 2-term Zeckendorf number: Zeckendorf decomposition uses exactly two Fibonacci terms: n = F_a + F_b with a ≥ b+2.
  Details: 42 = 34 + 8 (F9 + F6)
- Sparse Zeckendorf representation: Zeckendorf decomposition uses few Fibonacci terms (weight ≤ K).
  Details: weight 2 ≤ 3: 42 = 34 + 8 (F9 + F6)

Polygonal and Figurative Numbers
- Pronic number: A number of the form n(n+1), the product of two consecutive integers.
  Details: 42 = 6 × 7

Profile: default - Enter an integer, command or profile (h=Help, q=Quit): |
```

Figure 1 Example of NumClass output for the input $2^{**}5+10$, resulting in number 42.

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Introduction

NumClass is a modular and extensible software package written in Python, designed to classify integers into 11 main categories and over 200 distinct number types. The system covers a wide range of mathematical properties, including:

- **Arithmetic and Divisor-based** such as Perfect, Abundant, Deficient, Powerful, Practical, Sociable, Triperfect, and more.
- **Combinatorial and Geometric** such as Bell, cake numbers, Catalan, Motzkin, Stirling, Ramsey, etc.
- **Conjectures and Equation-based** including Egyptian fractions, Erdős–Straus, Goldbach, Legendre, and others.
- **Digit-based** like Palindrome, Automorphic, Happy, Narcissistic, Self, Kaprekar, etc.
- **Diophantine Representations** Numbers expressible as sums of squares or cubes.
- **Dynamical Sequences** such as Collatz, Ducci, Fibonacci mod n and Kaprekar routines.
- **Fun Numbers** those notable in pop culture, internet, computing, or science fiction.
- **Mathematical Curiosities** including, Munchausen, Kaprekar constants, Strobogrammatic, Vampire and Eban numbers.
- **Named Sequences** such as Busy Beaver, Carol, Lucky, Keith, Padovan and Taxicab numbers.
- **Polygonal and Figurate** including Triangular, Pentagonal, Pronic, Cyclic, and Repunit numbers.
- **Primes and Prime-related numbers** including twin, safe, sexy, Sophie Germain, palindromic, emirp, factorial, strong, super primes, and more.
- **Pseudoprimes and Cryptographic numbers** such as Blum, Carmichael, Fermat, and Euler–Jacobi pseudoprimes.

For a complete list of classifiers and detailed explanations, see Appendix A.

NumClass provides comprehensive number and divisor statistics, efficient logic for intersecting properties (e.g., “happy palindromic prime”), and explanations for each classification. It leverages OEIS¹ b-files where speed matters, automatically discovers custom classifiers and settings, and adapts its output to your terminal size.

This manual describes NumClass version 2.0, the manual itself is versioned independently. Manual version 1.0 corresponds to the initial public release of NumClass 2.0. Future manual revisions may document later NumClass releases or clarify existing functionality without changing the software version.

Manual Version	Date	NumClass Version	Notes
1.1	July 2026	2.2	Added features & refactoring
1.01	February 2026	2.0	Formatting, textual improvements
1.0	January 026	2.0	Initial manual release

¹ OEIS Foundation Inc. (2026), The On-Line Encyclopedia of Integer Sequences, Published electronically at <https://oeis.org>

Installation

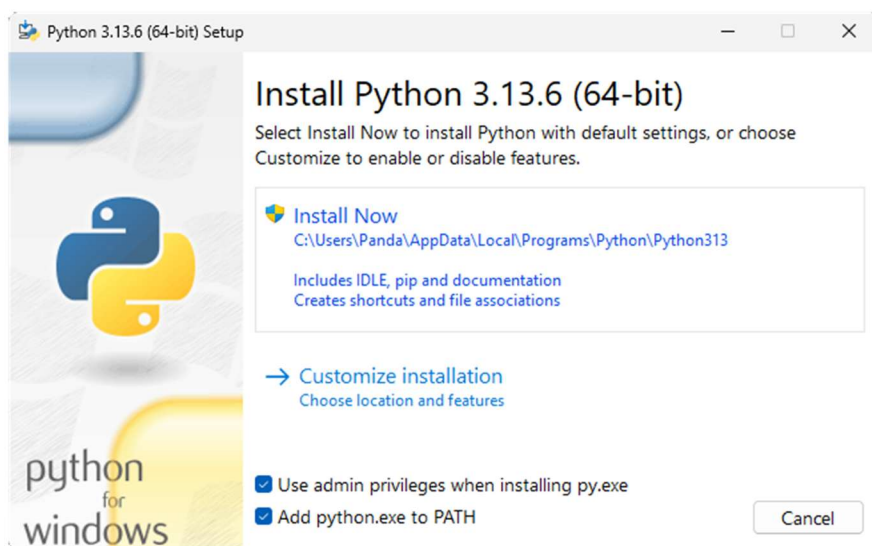
Prerequisites

- **Operating System:** Windows 10/11, macOS 12+, or any recent 64-bit Linux distribution
- **CPU:** Any 64-bit processor (e.g., Intel i5, Ryzen 3 or better)
- **RAM:** 8 GB or more recommended
- **Disk Space:** 1 GB free (for virtual environment and caches; SSD recommended)
- **Python:** Version 3.11 or higher (64-bit)
- **Terminal:** UTF-8 capable (for proper display of mathematical symbols)

Install Python

Windows

1. Download the latest Python release from python.org/downloads/windows.
2. Run the installer and check both boxes at the bottom:
 - “Use admin privileges when installing py.exe”
 - “Add python.exe to PATH”
3. Click “Install Now” to complete the installation.



Linux

- Most modern Linux distributions include Python 3 by default. To check your version, run:
`python3 --version`
- If you need to install or upgrade Python 3.11+:
 - **Debian/Ubuntu:**
`sudo apt update`
`sudo apt install python3 python3-pip`
For the latest Python, you may need to use a PPA or build from source.
 - **Fedora:**
`sudo dnf install python3 python3-pip`
 - **Arch Linux:**
`sudo pacman -S python python-pip`

macOS

- macOS 12+ may include Python 3, but it is recommended to install the latest version via Homebrew:
`brew install python`
- Verify installation:
`python3 --version`

Install pipx

pipx allows you to install and run Python applications in isolated environments.

Windows

```
py -m pip install --user pipx
py -m pipx ensurepath
```

Restart your terminal after running ensurepath.

Linux/macOS

```
python3 -m pip install --user pipx
python3 -m pipx ensurepath
```

Open a new terminal window, or run `source ~/.bashrc`, `source ~/.zshrc`, or the relevant shell configuration file to apply the changes immediately. A full restart is usually not necessary.

Install NumClass

Windows

```
pipx install "git+https://github.com/c788630/Numclass@main"
```

Linux / macOS

```
python3 -m pipx install "git+https://github.com/c788630/Numclass@main"
```

iOS

Numclass can run on iOS and iPadOS using the a-Shell app, available from the App Store. a-Shell provides a local Python environment, but it does not support git, so Numclass must be installed from a downloaded ZIP file.

To download the Numclass ZIP file on the iPad, open the Numclass GitHub release page in Chrome or Safari and select the latest release (for example, NumClass-2.2.0)

Scroll down to the **Assets** section and tap Source code (zip).

Tap Save at the bottom windows and select **Files** in the Save to...window.

Open the **Chrome** folder (or the folder used by your browser) to verify that the file numclass-2.2.0.zip has been downloaded.

Tap the ZIP file in the Files app. iOS will automatically extract the archive and create a folder named:

```
Numclass-2.2.0
```

The a-Shell documents folder may not be immediately visible in the iOS Files app if it does not yet contain any files. To force iOS to create and display the a-Shell folder, open the a-Shell app and run:

```
echo test > test.txt
```

Now open the Files app and go to: On My iPad.

The a-Shell folder should now be visible.

Open the Chrome folder, tap and hold the numclass-2.2.0 folder and select Copy.

Navigate to the a-Shell folder and tap and hold on an empty area, and select Paste.

Open the a_Shell app and enter the folder:

```
cd numclass-2.2.0
```

Install Numclass with pip:

```
python -m pip install .
```

Notes for iOS / iPadOS

Numclass uses a special safe factoring backend on iOS.

Because iOS/a-Shell does not safely support multiprocessing-based hard timeouts, Numclass disables subprocess factorization on this platform.

This means:

- Numclass itself runs normally.
- Large factorizations may be incomplete.

SteamOS

(open konsole)

```
set password: passwd [password]
```

```
sudo steamos-readonly disable
```

(enter password)

```
sudo pacman-key --init
```

```
sudo pacman-key --populate archlinux
```

```
sudo pacman -Sy python-pipx
```

```
check: pipx --version
```

```
pipx ensurepath
```

(restart terminal)

```
pipx install "git+https://github.com/c788630/Numclass@main"
```

First-Time Setup

After installation, initialize your workspace:

```
numclass init
```

This creates a NumClass workspace folder in your user Documents folder with classifiers, data, and profile files.

The paths used can be shown / verified with:

```
numclass where
```

Usage

After installation, you can use NumClass from the command line to classify integers and explore their mathematical properties.

Basic Command Structure

```
numclass [profile] [integer] [options]
numclass -h | --help
numclass init [overwrite]
numclass list
numclass where
```

numclass without an integer specified will start interactive mode.

[profile]	Starts NumClass interactive mode with the specified profile.
[integer]	The Integer or mathematical expression to classify.
[profile integer]	Specify a profile to use for this classification followed by the Integer or mathematical expression to classify. The profile is used for this run only.

If both a profile and an integer are provided, the profile must appear first.

Options:

-h, --help	Show help information and exit.
--output OUTPUT	Write results to a file (also prints to screen unless --quiet is used).
--quiet	Suppress screen output.
--no-details	Do not show explanations/details.
--debug	Enable debug mode for detailed timing, error and discovery information.

Commands:

active	Show the active profile.
init	Create workspace folders and copy packaged sample files.
init overwrite	Meant for developers. Requires environment variable <code>NUMCLASS_DEV=1</code> . Copies all initial classifiers, datafiles and profiles. All files in Documents/Numclass will be overwritten.
list	List all available classifiers.
where	Show the workspace and package paths.

Examples

`numclass`

Running `numclass` without arguments starts interactive mode. You will be prompted to enter:

- An integer, e.g. 42
- A mathematical expression, e.g. $2^{100}-1$
- One of the following commands:

<code>debug on off status</code>	to switch debug mode on, off or show current debug status.
<code>fast on off status</code>	to switch fast mode on, off or show current fast status.
<code>h or help</code>	to enter Help mode.
<code>hist</code>	show a history of the entered numbers.
<code>p</code>	to list available profiles and show the active profile.
<code>q or quit</code>	to quit NumClass.
- A profile name, like 'default' to switch between profiles.

`numclass 42`

Classifies number 42 using the last active profile.

`numclass 2**100-1`

Converts mathematical expression $2^{100}-1$ to 1267650600228229401496703205375, prints the results and exits.

`numclass aliquot 2856`

Classifies 2856 using the 'aliquot' profile (for this run only).

Help Mode

Type 'h' at the input prompt to open *help mode*.

This displays an introduction to NumClass, after which you can enter any of the following help subcommands:

- s - List effective settings**
Effective settings are auto discovered from Python code and the file: default.toml
Value legend in terminal output: white: default, yellow: set in profile, red: not used in Python code.
(Paginated output, [Enter] to continue, 'q' to quit help mode.)
- c - List all classifications by category**
Lists all available classifiers, grouped by category with a compact description of its function. (Paginated output, [Enter] to continue, 'q' to quit help mode.)
- r - Show OEIS references**
References to the Online Encyclopedia of Integer Sequences. Ctrl+Click the number to follow the link.
(Paginated output, [Enter] to continue, 'q' to quit help mode.)
- e - Show some example numbers you can try**
Some example numbers you can try.
Tip: select and copy the number, quit help mode and paste the value.
- l - Show symbols and notation legend**
Explains symbols and shorthands used in NumClass output.
- q - Quit help mode and start classifying integers**
Quit help mode and return to the input prompt.

Working With Profiles

Profiles are a core concept in NumClass. A profile defines how NumClass behaves, including which classifiers are active, which categories are enabled, performance limits, output settings, and display options.

Using profiles allows you to switch between different working modes (for example fast testing, full classification, experimental classifiers, or custom setups) without modifying code or command-line arguments.

What a Profile Controls

A profile can control, among others:

- Enabled and disabled classification categories
- Performance limits and time budgets
- Debug and fast modes
- Output behavior (OUTPUT_FILE)
- Display options
- Feature flags

Profiles make NumClass highly configurable while keeping the command line simple.

Temporary and Persistent Profile Selection

NumClass distinguishes between temporary profile selection (single-run) and persistent profile switching (interactive mode).

Command-line Profile Selection (Temporary)

When a profile name is specified together with a number on the command line, the profile is used only for that single run.

Example:

```
numclass collatz 360
```

Behavior:

- The collatz profile is used for this classification only
- The previously active default profile is not changed
- The next NumClass start will reuse the original default profile

This allows testing different profiles without modifying your persistent configuration.

Interactive Profile Switching (Persistent)

When NumClass is started in interactive mode (without providing a number), entering a profile name switches the active profile and makes it the new default.

Example:

```
numclass
```

Then at the prompt: `primes`

Behavior:

- The active profile is changed to primes
- The new profile is stored
- Future NumClass sessions will automatically start using this profile

NumClass persists the most recently selected interactive profile by writing its name to file: numclass\profiles\current. Each profile is stored as a TOML configuration file.

Listing Available Profiles

In interactive mode enter p to show a list of available profile files, stored in the /profiles directory. The currently active profile is marked with an arrow.

The following profiles are initially delivered with NumClass:

aliquot	Focus on aliquot theory: σ/τ stats, aliquot/quasi/unitary chains, peaks and timing.
all	Enable every category/classifier; useful for demos and maximal coverage.
collatz	Explore Collatz: sequence output, stopping time, peaks, optional extended step caps.
combinatorial	Combining classical counting sequences: Bell, Stirling, Fubini, Lah, Catalan, Motzkin, and related combinatorial numbers.
conjecture	Turn on conjectures: Goldbach/Legendre/Lemoine etc., witness listing; may be slow.
core	Classical number theory focus; no recreational/dynamic/curiosity.
default	Balanced everyday setup: broad coverage, safe caps, informative details.
lean	Fast minimal profile: headline classifiers only, terse output, strict caps.
primes	Primes, prime related and Pseudoprimes only.
sequences	Dynamical- and Named sequence based only.
statonly	Display the number statistics only.

Creating and Editing Profiles

Profiles are simple text files and can be edited with any text editor.

To create a new profile:

- Copy an existing profile file (best to use all.toml as it contains all settings)
- Rename it
- Modify the settings as needed

Go to your Documents/Numclass/profiles folder, edit a profile file e.g. default.toml change or add any setting and save the file.

Deleting a Profile

Go to your Documents/Numclass/profiles folder and simply delete the file having the profile name you want to delete.

Input Formats

NumClass accepts:

- Positive or negative integers (with optional thousand separators: spaces, underscores, periods or commas).
- Binary (0b), octal (0o) and hexadecimal (0x) numbers.
- Mathematical expressions that evaluate to integers.
- Special input translations as defined in Special_inputs.tsv.
This mechanism allows NumClass to support rich, interactive input beyond plain integers.

Mathematical Expressions: Operators and Precedence

Mathematical expressions resulting in an integer are allowed.

Example: 2^{100-1} (2 to the power of 100 minus 1) will be converted into 1267650600228229401496703205375.

Allowed operations and precedence. (x represents a number)

Prec.	Operators	Example(s)	Meaning
1	()	$(2 + 3) * 4$	Explicit grouping (highest precedence)
2	!x, x!!, x!	$5! \rightarrow 120$	Prefix subfactorial, postfix double factorial, and postfix factorial. Factorial operators bind tighter than **
3	**	$2 ** 10 \rightarrow 1024$	Exponentiation (right associative). Negative exponents not allowed.
4	+x, -x	-5, +8	Unary sign operators
5	*, //, %	$6 * 7$, $17 // 3$, $17 \% 3$	Multiplication, integer division, modulo
6	+, -	$7 - 4$, $2 + 3$	Addition and subtraction
7	<<, >>	$3 << 4$, $100 >> 3$	Bit shifts (shift count ≥ 0)
8	&	$42 \& 15 \rightarrow 10$	Bitwise AND
9	^	$5 \wedge 12 \rightarrow 9$	Bitwise XOR
10		$5 12 \rightarrow 13$	Bitwise OR
11	:	$12:34 \rightarrow 1234$	Decimal concatenation (lowest precedence)
12	?, ?n, ?(expr)	?100 $\rightarrow$ Random number between 0-100 ?500-250 $\rightarrow$ Random number between -250 and +250.	Random number placeholder, replaced before evaluation. ? generates a value in 0-999999 ?n generates a value in 0 – n. ?(expr) evaluates the expression and uses the result as the upper bound.

NumClass can work with Integers having up to 100000 digits by default.

This value can be changed using environment variable PYTHONINTMAXSTRDIGITS.

All known (2025) Mersenne prime expressions are handled even if the number of digits exceeds 100000, but the output is reduced to only show a few elements.

e.g. $2^{136279841-1}$ is a Mersenne prime and has 41024320 digits.

Examples of valid input:

42 positive integer.
-7 negative integer.
123 456 789 123456789 (using ' ' as thousand separator).
123.456.789 123456789 (using . as thousand separator).
123,456,789 123456789 (using , as thousand separator).

1_000_000	1000000 (using _ as thousand separator).
0b1010	10 (using binary prefix 0b).
0o77	63 (using octal prefix 0o).
0xFF	255 (using hexadecimal prefix 0x).
2**1000-1	1267650600228229401496703205375

Invalid input:

3.14	decimal point not allowed as it is not an integer and not a thousand separator.
1,23	comma not used as a thousand separator (requires groups of 3 digits).
12.34.56	period not used as a thousand separator (requires groups of 3 digits).
0xG1	invalid value after hexadecimal prefix (only digits 0-9, letters A-F are allowed).
0o80	invalid value after octal prefix (only digits 0-7 are allowed).
3+2i	rejected as a complex number is not an integer by definition.
(2**10)/3	rejected as non-integer division is not allowed (would create a float 141.333...)

Input Redirection

NumClass supports standard input redirection on Windows, Linux, and macOS. This allows numbers to be read from a file or pipe instead of being entered on the command line.

This feature can be used both for very large single integers and for batch processing multiple numbers.

```
numclass --output results.txt < input.txt
```

This method is intended for processing extremely large numbers that exceed practical command-line length limits.

The redirected input file may contain one or multiple integer values or mathematical expressions, separated by line breaks.

Example input.txt:

```
42
36056789012345678901234567890
2**100-1
```

Each line is treated as a separate NumClass input. Leading and trailing whitespace is ignored.

Batch processing behavior

When multiple input values are provided via redirection:

- Each number is processed sequentially
- Output is generated for each input
- When using file output, results are appended or written per-number depending on the selected output mode

Batch processing is most useful when writing output to files (--output), since interactive screen output is cleared between runs.

Command-line input precedence

If both a number/expression is specified on the command line and standard input is redirected, then the command-line number takes precedence and the redirected input is ignored.

User-defined input transforms

Numclass can transform non-decimal textual input into integers before classification.

Custom input transforms can be added by placing a Python file in:

Numclass/inputs/

Built-in examples are:

- Roman numerals, e.g. *MCMLXVIII* = 1968 (*roman.py*)
- Phonetic Klingon numbers e.g. *javSaD wa'vatlh SochmaH loS* = 6174 (*klingon.py*)

Each transform file must define a function named:

```
def transform(text: str) -> int:
```

The function must follow this rule:

- * If the complete input is recognized, return an integer.
- * If the input does not belong to this transform, raise `TransformNotApplicable``.
- * If the input looks like this transform but is invalid, raise `ValueError``.

Transforms must parse the complete input. They should never extract only part of the text.

For example, a transform may accept:

foo:42

but must not silently accept:

2*foo:42

Minimal template

```
from numclass.transform import TransformNotApplicable

def transform(text: str) -> int:
    """
    Convert custom textual input to an integer.

    Return:
        int

    Raise:
        TransformNotApplicable
            if this transform does not apply to the input.

        ValueError
            if the input looks like this transform, but is invalid.
    """

    s = (text or "").strip()
    if not s:
        raise TransformNotApplicable

    # Example syntax: my:123
    if not s.lower().startswith("my:"):
        raise TransformNotApplicable
```

```
value = s[3:].strip()

if not value:
    raise ValueError("Missing value after 'my:'.")

if not value.isdigit():
    raise ValueError(f"Invalid custom input: {text!r}")

return int(value)
```

Important

Transforms are full-input parsers, not search functions.

Do this:

```
if not s.startswith("my:"):
    raise TransformNotApplicable
```

Do not do this:

```
if "my:" in s:
    ...
```

The second form can accidentally transform expressions or unrelated input.

Output Files

NumClass can write classification results either to individual output files (one file per number) or append multiple results to a single file. Output destinations can be specified using the profile setting `OUTPUT_FILE` or via the command-line option `--output`. Both methods use identical path resolution and filename handling rules.

Example (command line):

```
numclass --output C:\NumClass\results
```

Basic Output Configuration

Disable file output:

```
OUTPUT_FILE = ""
```

No output file is created. Results are displayed only in the terminal.

Per-number output (directory mode)

When `OUTPUT_FILE` ends with a directory separator (/ or \), NumClass creates one output file per evaluated number.

```
OUTPUT_FILE = "results/"
```

Result: `<WORKSPACE>/results/<n>.txt`

Example: 42 → `<WORKSPACE>/results/42.txt`

Workspace root shortcut

```
OUTPUT_FILE = "."
```

Writes per-number output files directly into the NumClass Workspace root:

`<WORKSPACE>/<n>.txt`

Note: `"."` refers to the NumClass Workspace directory, not the current shell directory.

Single File Output (Append Mode)

When `OUTPUT_FILE` specifies a file path (not ending with /), NumClass appends all classification results to the same file.

```
OUTPUT_FILE = "log.txt"
```

Result: `<WORKSPACE>/log.txt`

Absolute file paths may also be used:

```
OUTPUT_FILE = "/data/numclass.log"
```

```
OUTPUT_FILE = "C:\Logs\numclass.txt"
```

Path Resolution Rules

NumClass resolves output paths using the following rules.

Home directory expansion (~)

Paths starting with `~` are expanded by NumClass to the user's home directory (Linux, macOS and Windows supported):


```
OUTPUT_FILE = "~/numclass/results/"
```

- C:\Users\<username>\numclass\results (Windows)
- /home/user/numclass/results (Linux/MacOS)

Relative paths

All relative paths ending with “/” are interpreted relative to the NumClass Workspace directory:

```
OUTPUT_FILE = "results/"
```

Result: <WORKSPACE>/results/<n>.txt

Absolute paths

Absolute paths are used unchanged:

```
OUTPUT_FILE = /data/results
OUTPUT_FILE = C:\NumClass\results
```

Workspace location

To display the location of the active workspace directory, use:

```
numclass where
```

Automatic Filename Shortening for Very Large Numbers

When writing per-number output files, NumClass normally uses the evaluated number itself as filename:

```
42.txt
-991.txt
```

However, operating systems impose limits on filename and full path length. Since NumClass supports integers with up to 100000 digits, this limit can be exceeded.

To ensure reliable cross-platform operation, NumClass automatically switches to a shortened, filesystem-safe filename format when the full output path would exceed approximately 250 characters.

Shortened filename format:

```
digits=<D>_<SIGN><FIRST12>...<LAST12>_sha=<HASH>.txt
```

Where:

Field	Description
D	Total number of decimal digits (excluding the minus sign)
SIGN	- for negative numbers, empty for positive numbers
FIRST12	First 12 digits of the number
LAST12	Last 12 digits of the number
HASH	First 12 hexadecimal characters of the SHA-256 hash of the full decimal value

The hash component is used as a compact, collision-resistant identifier. While cryptographic collisions are theoretically possible, the combined filename structure makes accidental clashes practically negligible.

Large number example:

digits=100000_314159265358...897932384626_sha=73475cb40a56.txt

For small numbers where the full path length remains within limits, the original filename format is preserved, e.g. 42.txt

Versioned Output Files (No Overwrite)

If a target filename already exists, a numeric suffix is appended before the file extension:

Example:

42.txt

42_2.txt

42_3.txt

The initial output file uses the base filename without a suffix. Additional versions start at _2. This behavior also applies to shortened filenames for very large numbers.

Example (large-number versioning):

digits=100000_314159265358...897932384626_sha=73475cb40a56.txt

digits=100000_314159265358...897932384626_sha=73475cb40a56_2.txt

This design is intentional. Re-running NumClass may produce different results (for example due to warmed caches or different effective limits). Automatic versioning prevents loss of earlier output.

Number and Divisor Statistics

When NumClass evaluates an integer, it first prints a **Number statistics** section, followed by **Divisor statistics** (of $|n|$).

These statistics provide fast, fundamental information about the number before the detailed classification results.

Number statistics

This section summarizes basic arithmetic, digital, and multiplicative properties of the input number.

Input

If the entered value is an expression that evaluates to an integer, NumClass displays the original input expression in addition to the evaluated number.

This provides transparency about how the final number was obtained.

Example:

```
Input:  2**5+10
Number: 42
```

If the input is already a literal integer, the Input line is omitted.

Number

The evaluated integer value. This is either the number as entered (which may be negative), or the result of the input expression.

Digits

- *Count*: Number of decimal digits of $|n|$
- *Sum*: Sum of the decimal digits
- *Product*: Product of the decimal digits

Parity

Indicates whether the number is *Even* or *Odd*.

Prime

Indicates whether the number is prime.

For composite numbers, the nearest smaller and larger prime numbers are shown.

Additionally, for numbers up to 10^{10} , NumClass displays the **prime-counting function** $\pi(n)$, the number of primes less than or equal to n , and the **prime density**, the percentage of integers from 1 through n that are prime. These statistics are omitted for larger numbers to maintain performance.

Digital root of $|n|$

The repeated sum of decimal digits until a single digit is obtained.

Also shows:

- the reduction sequence
- the additive persistence (number of steps)

Multiplicative persistence

The number of steps required to reduce $|n|$ to a single digit by repeatedly multiplying its decimal digits. The full reduction sequence is shown.

Zeckendorf

The *Zeckendorf decomposition* of the number: a unique representation as a sum of *non-consecutive Fibonacci numbers*.

Displayed as:

- the Fibonacci sum
- the corresponding Fibonacci indices F_k
- a compact Fibonacci-base bitstring

The *Fibonacci-base bitstring* represents the Zeckendorf decomposition in Fibonacci base: each bit corresponds to a Fibonacci number F_k , ordered from the largest index down to F_2 .

- a bit value **1** indicates that F_k is present in the decomposition,
- a bit value **0** indicates that it is not,
- no two consecutive bits are 1, reflecting the Zeckendorf rule.

Example:

$42 = 34 + 8$ ($F_9 + F_6$), $\text{bits}(F_9..F_2)=10010000$

Prime factorization

The canonical factorization of $|n|$ into prime factors.

Prime factor counts

- $\Omega(n)$: total number of prime factors (with multiplicity)
- $\omega(n)$: number of distinct prime factors

Euler's totient $\phi(n)$

The number of integers $1 \leq k \leq n$ that are coprime to n .

The calculation is shown using multiplicativity over the prime factors.

Carmichael function $\lambda(n)$

The exponent of the multiplicative group modulo n .

Computed as the least common multiple of the individual prime-power λ -values.

1/n (base 10)

Decimal expansion of the reciprocal of n , including:

- repeating or terminating behavior
- length of the repeating period
- length of the preperiod (if any)

Möbius $\mu(n)$

The Möbius function value:

- 0 if n is not squarefree
- ± 1 if n is squarefree (sign determined by the parity of prime factors)

Radical $\text{rad}(n)$

The product of the distinct prime divisors of n .

Divisor statistics of $|n|$

This section summarizes properties derived from the *divisors of $|n|$* .

Divisor count $\tau(n)$

The total number of positive divisors of n .

Divisors

The complete list of positive divisors, in increasing order.

Classical divisor sums

- $\sigma(n)$: sum of all positive divisors
- $s(n)$: sum of proper divisors ($\sigma(n) - n$)
- $q(n)$: $s(n) - 1$

Unitary divisor sums

- $\sigma^*(n)$: sum of unitary divisors
- $s^*(n)$: sum of proper unitary divisors
- $\tau^*(n)$: number of unitary divisors
- $q^*(n)$: $s^*(n) - 1$

Aliquot sequence

The sequence obtained by repeatedly replacing the number with the sum of its proper divisors.

Displayed with:

- full sequence
- number of steps
- peak value and its position
- computation time

Quasi-aliquot sequence

A variant aliquot sequence using $q(n)$ instead of $s(n)$, shown with the same statistics.

Notes

- All divisor statistics are computed for $|n|$.
- Some statistics may be omitted or abbreviated in *fast mode*.
- Expensive computations are guarded by time and size limits to keep NumClass responsive.

Advanced Usage

Developer Mode

To enable developer features, set the environment variable:

```
set NUMCLASS_DEV=1      (Windows)
export NUMCLASS_DEV=1   (Linux/macOS)
```

Currently this enables the `numclass init overwrite` command to overwrite the classifiers, data files and profiles stored in the Documents\Numclass folder with the ones present in the NumClass package folders. Any changes made to the files in the Documents\Numclass folder will be lost!

Creating a Classifier

NumClass comes with 200+ classifiers, this section is only needed if you want to add your own classifier.

1. Each classifier starts with a decorator:

```
@classifier(
    label="<your classifier label>",
    description="<your classifier description>",
    oeis="<oeis reference>",      # Optional
    category="<existing or new category>",
    limit="<your integer limit>" # Optional
)
```

2. Define a function starting with “is_”. This prefix is needed so the function is auto discovered by NumClass:

```
Def is_<your function name>(n: int, ctx: NumCtx | None = None) ->
tuple[bool, str | None]:
```

ctx (context) is optional and can be used if your function uses precomputed factor-based data, When present, ctx is an instance of the following dataclass:

```
@dataclass(frozen=True)
class NumCtx:
    n: int
    fac: dict[int, int]      # May include composite cofactors if factoring
                             hit limits.
    sigma: int | None        #  $\sigma(n)$ ; None if incomplete or  $n$  in  $\{0\}$ 
    tau: int | None          #  $\tau(n)$ ; None if incomplete or  $n$  in  $\{0\}$ 
    unitary_sigma: int | None #  $\sigma^*(n)$ ; None if incomplete or  $n$  in  $\{0\}$ 
    unitary_tau: int | None   #  $\tau^*(n)$ ; None if incomplete or  $n$  in  $\{0\}$ 
    omega: int | None        # Total prime factors with multiplicity;
                             None if incomplete or  $n$  in  $\{0\}$ 
    incomplete: bool = False # True if fac contains any composite base.
    composite_bases: tuple[int, ...] = () # Optional: composite bases,
                                         useful for UI/debug.

@property
def is_fully_factored(self) -> bool:
    return not self.incomplete
```

3. Perform your classification logic and return either `True`, "your details" or `False`, `None`
4. Specifying `False`, "your details", is also allowed; those details are only shown in debug mode.

Classifier example

```
from numclass.utility import build_ctx,    ← if you want to use factor based data

@classifier(
    label="Sphenic number",
    description="A positive integer that is the product of three distinct primes.",
    oeis="A007304",
    category="Arithmetic and Divisor-based",
    limit=999_999_999,
)
def is_sphenic_number(n: int, ctx: NumCtx | None = None) -> tuple[bool, str | None]:
    """
    Returns (True, details) if n is a product of 3 distinct primes,
    else (False, None).
    """
    if n < 30: # smallest sphenic number is 2*3*5 = 30
        return False, None

    # Reuse a precomputed context if provided, otherwise build a fresh one.
    ctx = ctx or build_ctx(n)
    f = ctx.fac

    if len(f) == 3 and all(exp == 1 for exp in f.values()):
        primes = list(f.keys())
        return True, f"{n} = {primes[0]} × {primes[1]} × {primes[2]}"

    return False, None
```

Using profile settings in your classifier

Import the CFG function:

```
from numclass.runtime import CFG
```

To read a setting from the active profile, e.g. `FAST_MODE` in section [BEHAVIOUR] use:

```
Fast_mode = CFG("BEHAVIOUR.FAST_MODE", False)
```

It returns the specified setting or the second CFG parameter as a default value if not specified in the active profile.

Using OEIS b-files

NumClass can use precomputed integer sequences from the [OEIS](#) to accelerate classification and avoid recomputation of known results.

These sequences are stored in so-called b-files, which list each term of a sequence in the form:

```
# A000110 Bell numbers example
0 1
1 1
2 2
3 5
4 15
...
```

To access these efficiently, NumClass provides:

```
from numclass.utility import check_oeis_bfile
```

Example

To check whether a number occurs in the Highly abundant numbers (OEIS A002093):

```
found, idx_file, series, idx_set = check_oeis_bfile("b002093.txt", n)
```

This returns a 4-tuple:

Return name	Type	Meaning
found	bool	True if n appears in the sequence
idx_file	int None	Index from the first column of the b-file (usually the OEIS term number)
series	list[int]	All sequence values in file order
idx_set	int None	0-based position of n inside series (useful for retrieving neighbors)

Example usage:

```
if found:
    print(f"{n} is term {idx_file} in A002093.")
    if idx_set > 0:
        prev = series[idx_set - 1]
        print(f"Previous record: {prev}")
```

You can use `series[idx_set - 1]` to access the previous value in the sequence if `idx_set > 0`.

Resolution order

When you call `check_oeis_bfile(name, n)`, NumClass automatically looks for the file in:

1. **Your workspace** — `<workspace>/data/<name>`
2. **The packaged defaults** — `numclass.data/<name>`

Notes

- `found` → False and empty series means the file was not found or failed to load.

Data Files Used by NumClass

NumClass uses the following special data files present in folder `\data`:

Besides OEIS *b-files*, NumClass uses several small, curated data files to recognize special numbers, provide fast lookups, define combined classifications, and support Easter eggs and demonstrations.

All data files share these properties:

- Human-readable
- Safe to edit
- Reloaded on restart
- No Python execution
- Workspace overrides packaged defaults
- Comment lines (starting with #) and empty lines are allowed and ignored

If a data file contains errors (wrong format, missing fields), NumClass reports a clear, user-friendly error message.

curiosity_constants.tsv

Purpose: Defines famous mathematical constants that are single values or small finite sets, not infinite sequences.

Used by: Curiosity constants classifier (data-driven)

Typical entries include: Kaprekar constants, Digit-Reversal Constant, Harshad in all bases, etc.

Format:

label<TAB>n<TAB>description<TAB>details<TAB>oeis

Example:

Harshad in all bases<TAB>1,2,4,6<TAB>An integer that is divisible by the sum of its digits in every base ≥ 2 .<TAB>Only 1, 2, 4, and 6 are Harshad in every base ≥ 2 .<TAB>

This design allows NumClass to generate many classifiers automatically from data alone.

erdos_woods.toml

Purpose: Provides a complete known list of Erdős–Woods numbers and auxiliary data for fast classification.

Used by: Erdős–Woods number classifier is_erdos_woods_number() in named_sequences.py

Structure:

- [erdos_woods] — known values
- [erdos_woods.min_starts] — minimal starting points for verification

This avoids expensive recomputation and ensures correctness for large inputs.

fun_numbers.tsv

Purpose: Defines “fun numbers” from culture, science, computing, memes, and popular mathematics. This file allows users to add their own trivia without touching Python code.

Used by: Fun number classifier

Format: number<TAB>description

- Descriptions are free text
- Numbers may be negative

Example:

42<TAB>The Answer to Life, the Universe, and Everything.

Graham_last_digits.txt

Contains the last 10000 digits of Grham’s Number in rows of 100 digits.

hard_factors.txt

Purpose: Stores known factorizations of very large or computationally expensive integers, including:

- Fermat numbers
- RSA challenge numbers
- Difficult aliquot sequence values

Used by: Factorization routines and classifiers that depend on divisor structure in utility.py

Format:

Whitespace-separated values (number followed by known factors or expressions).

This file dramatically improves performance for known “hard” cases.

intersections.toml

Purpose: Defines combined classifications (“intersections”) when a number satisfies multiple atomic properties.

Used by: Intersection engine

Example:

```
[[intersections]]
of = ["Prime number", "Palindrome"]
label = "Palindromic prime"
category = "Primes and Prime-related Numbers"
```

Key points:

- Order does not matter
- More specific intersections take precedence
- Fully user-customizable
- No Python code required

This file allows users to extend NumClass's taxonomy declaratively.

special_inputs.tsv

Purpose: Handles non-numeric or symbolic inputs such as Easter eggs, shortcuts, and transformations.

Used by: Input parser (expreval.py)

Format: key<TAB>handler<TAB>index<TAB>description<TAB>digits<TAB>extra

```
2**136279841-1<TAB>mersenne_exact<TAB>52<TAB>Mersenne prime M52
2^136279841-1, discovered in 2024 by L Durant using GIMPS<TAB>41024320<TAB>
0**0<TAB>egg<TAB><TAB>00 is undefined – the true neutral element of
confusion.<TAB><TAB>
```

This mechanism allows NumClass to support rich, interactive input beyond plain integers.

sum_of_three_cubes.toml

Purpose: Contains known integer solutions to the equation: $x^3 + y^3 + z^3 = n$

Used by: Sum-of-three-cubes classifier

Structure:

```
[solutions]
33 = [8866128975287528, -8778405442862239, -2736111468807040]
42 = [-80538738812075974, 80435758145817515, 12602123297335631]
```

This allows NumClass to instantly recognize celebrity solutions without recomputation.

Troubleshooting

Start `numclass --debug` to enable debug mode. The following debug information will be shown:

Some runtime information:

```
[runtime] Platform: Windows-11-10.0.26200-SP0
[runtime] Python: 3.13.6 (win32, 64-bit)
[runtime] Terminal size: 149x83
[runtime] gmpy2 accelerator: enabled
```

Classifiers loaded in your workspace folder (ws):

```
[discovery] ws OK arithmetic_divisor.py: 52 label(s)
[discovery] ws OK cocombinatorial_geometric.py: 13 label(s)
[discovery] ws OK conjectures.py: 8 label(s)
[discovery] ws OK curiosities.py: 23 label(s)
[discovery] ws OK digit_based.py: 24 label(s)
[discovery] ws OK diophantine.py: 4 label(s)
[discovery] ws OK dynamical_sequences.py: 9 label(s)
[discovery] ws OK fun_number.py: 1 label(s)
[discovery] ws OK named_sequences.py: 19 label(s)
[discovery] ws OK polygonal_figurate.py: 11 label(s)
[discovery] ws OK prime.py: 36 label(s)
[discovery] ws OK pseudoprime_crypto.py: 8 label(s)
```

If your classifier contains an error, it will be listed here. Example:

```
[discovery] ws FAIL digit_based.py: NameError: name 'Decimal' is not defined
```

Classifier files already loaded from the workspace and found duplicate in the package folder will be skipped:

```
[discovery] SKIP 215 duplicate label(s) skipped.
```

The number of atomic classifiers and intersection rules loaded from file intersections.toml:

```
[discovery] Atomic classifiers discovered: 215
[discovery] intersections OK: 24 from
C:\Users\<user>\Documents\Numclass\data\intersections.toml
```

Number of input transformations loaded from input folder

```
[discovery] Input transformations loaded: numclass.inputs.klingon,
numclass.inputs.roman
```

The profile used and settings found in the loaded profile (partial example):

```
[debug] active profile: all
[debug] profile file: C:\Users\<user>\Documents\Numclass\profiles\all.toml
[debug] profile keys (raw → runtime value/type):
  _PROFILE.description..... <description> (str)
  _PROFILE.name..... 'all' (str)
  ALIQUOT.MAX_STEPS..... 250 (int)
  ALIQUOT.STEP_TIME_LIMIT..... 0.3 (float)
  ALIQUOT.VALUE_LIMIT..... 10000000000 (int)
  BEHAVIOUR.ALLOW_SLOW_CALCULATIONS..... False (bool)
  CATEGORIES.Arithmetic and Divisor-based..... True (bool)
  CATEGORIES.Combinatorial and Geometric..... True (bool)
```

Now the standard input prompt is displayed, allowing you to enter an integer or an expression.

Number Classifier v2.0.0 – Mathematical Classifications & Curiosities

Profile: all – Enter an integer or profile (H=Help, P=List profiles, Q=Quit):

After entering, for example, 42, the classification results are displayed along with the time taken to complete the classification.

```
[ 0.00 ms] NO   Highly composite number
[ 2.61 ms] NO   Superabundant number
[ 0.04 ms] NO   Superperfect number
[ 13.20 ms] NO   Untouchable number
[ 0.00 ms] NO   Weird number
[ 0.17 ms] OK   Egyptian fraction (K=5) –  $5/42 = 1/9 + 1/126$ 
```

```
[ 0.01 ms] OK    Egyptian fraction (K=6) -  $6/42 = 1/8 + 1/56$ 
[ 0.01 ms] OK    Erdős-Straus (K=4) -  $4/42 = 1/11 + 1/232 + 1/53592$ 
[ 0.02 ms] OK    Goldbach conjecture - 42 has 4 Goldbach pairs: 5+37, 11+31, 13+29, 19+23
(partial example)
```

If any errors occur during classification, they will also be displayed here, including the corresponding line number.

Further Resources

GitHub Repository: <https://github.com/c788630/Numclass>

Appendix A – Classifications and Categories

This appendix provides reference descriptions for all 11 NumClass categories. Each table lists each available classifier with a short description / mathematical definition, followed by an example, and OEIS reference.

Arithmetic and Divisor-based Numbers

The following categories contain divisibility- and divisor-function-based classifications.

Classifier	Description / Definition	Examples / OEIS reference
Abundant number	A number is <i>abundant</i> if the sum of its proper divisors is greater than the number itself. $s(n) > n$	20: Aliquot sum (sum of proper divisors): $s(20) = 1 + 2 + 4 + 5 + 10 = 22 (> 20)$ OEIS A005101
Achilles number	A <i>powerful number</i> that is not a perfect power; that is, all prime exponents in its factorization are ≥ 2 and their greatest common divisor equals 1.	72: $72 = 2^3 \times 3^2$. Both exponents ≥ 2 , but $\gcd(3, 2) = 1$, so 72 cannot be expressed as m^k with $m > 1$ and $k > 1$. OEIS A052486
Almost perfect number	A number is <i>almost perfect</i> if the sum of its divisors $\sigma(n) = 2n - 1$; only 1 and powers of 2 are known. $\sigma(n) = 2n - 1$	8: Sum of divisors: $\sigma(8) = 1 + 2 + 4 + 8 = 15 = 2 \times 8 - 1$ OEIS A000079
Amicable number	Two distinct numbers a and b are <i>amicable numbers</i> if each equals the aliquot sum of the other. $s(a) = b$ and $s(b) = a$.	220: Aliquot sum: $s(220) = 1 + 2 + 4 + 5 + 10 + 11 + 20 + 22 + 44 + 55 + 110 = 284$. $s(284) = 1 + 2 + 4 + 71 + 142 = 220$; 220 and 284 form an amicable pair since $s(220) = 284$ and $s(284) = 220$. OEIS A063990
Aspiring number	Number whose aliquot sequence eventually reaches a perfect number, though it is not perfect itself.	25: Aliquot sum: $s(25) = 1 + 5 = 6$, 6 is perfect since the sum of proper divisors $s(6) = 1 + 2 + 3 = 6$ $\Rightarrow 25$ is an aspiring number. OEIS A063769
Betrothed number (Quasi amicable number)	A <i>betrothed pair</i> is a pair of positive integers (a, b) such that: $s(a)=b+1$ and $s(b)=a+1$.	48: Aliquot sum: $s(48) = 1 + 2 + 3 + 4 + 6 + 8 + 12 + 16 + 24 = 76$. $s(75) = 1 + 3 + 5 + 15 + 25 = 49$; satisfies $s(48) = 75 + 1$ and $s(75) = 48 + 1$. (48,75) is a betrothed pair. OEIS A005276

Classifier	Description / Definition	Examples / OEIS reference
Colossally abundant number	A number is <i>colossally abundant</i> if it makes the ratio of the divisor sum to a slightly higher power of the number as large as possible for some positive ε , $\frac{\sigma(n)}{n^{1+\varepsilon}}$ is maximal for some $\varepsilon > 0$.	12: Sum of divisors: $\sigma(12) = 1 + 2 + 3 + 4 + 6 + 12 = 28$ and for an appropriate small ε , $\frac{\sigma(12)}{12^{1+\varepsilon}}$ exceeds that ratio for all smaller numbers. For a concrete witness, take $\varepsilon = 0.2$: $F_{12}(0.2) = 28 / 12^{1.2} \approx 1.4195$, while for $k = 1, \dots, 11$ the values are: 1.0000, 1.3058, 1.0703, 1.3263, 0.8697, 1.3977, 0.7744, 1.2370, 0.9308, 1.1357, 0.6753, all smaller than 1.4195. OEIS A004490
Cube-free number	A <i>cube-free</i> number has no prime factor raised to the third power or higher.	30: $30 = 2^1 \times 3^1 \times 5^1$ All prime factor exponents are ≤ 2 . OEIS A004709
Cube-full number	Integer where every prime exponent in its factorization is ≥ 3 .	216: $216 = 2^3 \times 3^3$ All prime factor exponents are ≥ 3 . OEIS A036966
Deficient number	Number whose proper divisors sum to less than the number. $s(n) < 2n$	16: Aliquot sum: $s(16) = 1 + 2 + 4 + 8 = 15 (< 16)$ OEIS A005100
Descartes number	A <i>Descartes number</i> is an odd positive integer satisfying $\sigma(n)=2n$ except that one composite factor is treated as if prime. 198585576189 is the only known Descartes number	198585576189: $198585576189 = 19 \times 43 \times 67 \times 1381$ (with 1381 composite) OEIS A174292
Friendly number	Two positive integers m and n are <i>friendly</i> if they share the same abundancy ratio, defined as $A(x) = \frac{\sigma(x)}{x}$ where $\sigma(x)$ is the sum of all divisors of x . In other words, m and n are friendly if $\frac{\sigma(m)}{m} = \frac{\sigma(n)}{n}$.	6: Sum of divisors: $\sigma(6) = 1 + 2 + 3 + 6 = 12 \Rightarrow A(6) = 12/6 = 2$ 28: $\sigma(28) = 1 + 2 + 4 + 7 + 14 + 28 = 56 \Rightarrow A(28) = 56/28 = 2 \Rightarrow$ 6 and 28 are friendly numbers. OEIS A074902
Giuga number	A <i>Giuga number</i> is a composite integer n such that for every prime divisor $p \mid n$, $p \mid \left(\frac{n}{p} - 1\right)$. In other words, when you divide n by each of its prime factors, subtract 1, the result must be divisible by that prime.	30: Prime divisors: $2 \times 3 \times 5$ Check each prime divisor $\frac{n}{p} = 15, 10, 6$: 15 - 1 is divisible by 2 10 - 1 is divisible by 3 6 - 1 is divisible by 5 All conditions are satisfied $\Rightarrow 30$ is a Giuga number. OEIS A007850

Classifier	Description / Definition	Examples / OEIS reference
Hemi-perfect number (half-perfect number)	A hemiperfect number is a positive integer whose abundancy index is a half-integer: $\frac{\sigma(n)}{n} = \frac{k}{2}$ for some odd integer k . Equivalently, $2\sigma(n)/n$ is an odd integer. All known hemiperfect numbers are even; the first few are 2, 24, 4320, 4680, 26208, ...	24: Sum of divisors: $\sigma(24) = 1 + 2 + 3 + 4 + 5 + 6 + 8 + 12 + 24 = 60$ Abundancy index: $\frac{\sigma(24)}{24} = \frac{60}{24} = \frac{5}{2}$, which is a half integer where 5 is odd $\Rightarrow$ 24 is a hemiperfect number. OEIS A159907
Hexaperfect number	A <i>hexaperfect number</i> (also called a 6-multiperfect number) is an integer whose sum of divisors equals six times the number itself. $\sigma(n) = 6n$	154345556085770649600: $\sigma(154345556085770649600) = 926073336514623897600 = 6 \times 154345556085770649600$ OEIS A046061
Highly abundant number	A <i>highly abundant number</i> is a positive integer n such that the sum of its divisors $\sigma(n)$ is greater than that of any smaller positive integer. In other words, n maximizes the divisor-sum function up to that point: $\sigma(n) > \sigma(k)$ for all $k < n$.	16: Sum of divisors: $\sigma(16) = 1 + 2 + 4 + 8 + 16 = 31$, among all integers $k < 16$, $\sigma(k) \leq 28$ the largest divisor sum is $\sigma(12) = 28$. Since $31 > 28$, 16 has the greatest divisor sum among all numbers $\leq 16 \Rightarrow$ 16 is highly abundant. OEIS A002093
Highly composite number	A <i>highly composite number</i> is a positive integer n that has more divisors than any smaller positive integer. That is, it maximizes the divisor-counting function $d(n)$ (also written $\tau(n)$) up to that point: $d(n) > d(k)$ for all $k < n$.	12: Divisors: 1, 2, 3, 4, 6, 12 $d(12) = 6$ For all smaller $k < 12$: $d(1) = 1, d(2) = 2, d(3) = 2, d(4) = 3, d(6) = 4, d(8) = 4, d(10) = 4$ None have more than 6 divisors $\Rightarrow$ 12 is highly composite. OEIS A002182
k-Hyperperfect number	A number n is k-hyperperfect if for some integer $k \geq 1$: $n = 1 + k(\sigma(n) - n - 1)$. Equivalently: $k = \frac{n-1}{\sigma(n) - n - 1}$. In other words: The “divisor excess” $\sigma(n) - n - 1$ must divide $n - 1$.	301: Sum of divisors: $\sigma(301) = 1 + 7 + 43 + 301 = 352$. $\sigma(n) - n - 1 = 352 - 301 - 1 = 50$ $(n - 1) / 50 = 300 / 50 = 6$ $k = 6$ is an integer $\Rightarrow$ 301 is a 6-hyperperfect number. OEIS A034897
k-Full number	Positive integer where every prime exponent is at least k (e.g., 4-full).	7776: $7776 = 2^5 \times 3^5$, $k = 5$ OEIS A036967 ($k = 4$)
Mersenne number	A <i>Mersenne number</i> is a positive integer of the form $n = 2^p - 1$ for some integer $p \geq 1$. Mersenne numbers arise naturally in number theory because powers of 2 play a central role in divisibility, binary representation, and perfect numbers.	2047: $2047 = 2^{11} - 1$ $2047 = 23 \times 89$. 2047 is a Mersenne number, but it is not prime.

Classifier	Description / Definition	Examples / OEIS reference
Near-perfect number	A positive integer n is <i>near-perfect</i> if n equals the sum of all its proper divisors except one.	12: Sum of proper divisors: $s(12) = 1 + 2 + 3 + 4 + 6 = 16$. If we omit $m = 4$, we get $1 + 2 + 3 + 6 = 12 \Rightarrow 12$ is near-perfect, with redundant divisor 4. OEIS A181595
Ore (harmonic divisor) number	A positive integer n is an Ore number if the harmonic mean of its positive divisors is an integer. Let the divisors of n be $d_1, d_2, \dots, d_k$. Then n is an Ore number if: $H(n) = \frac{k}{\sum_{i=1}^k \frac{1}{d_i}}$ is an integer. Using divisor functions, this can be written more compactly as: $H(n) = \frac{n \cdot \tau(n)}{\sigma(n)}$ and n is Ore if and only if $\frac{n \cdot \tau(n)}{\sigma(n)} \in \mathbb{Z}$, where: $\sigma(n)$ = sum of divisors $\tau(n)$ = number of divisors	6: Divisors = 1, 2, 3, 6 $H(6) = \frac{4}{\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{6}} = \frac{4}{2} = 2$ 2 is an integer $\Rightarrow 6$ is an Ore number. OEIS A001599
Pentaperfect number	A <i>pentaperfect number</i> (also called a 5-multiperfect number) is an integer whose sum of divisors equals five times the number itself. $\sigma(n) = 5n$.	14182439040: Sum of divisors: $\sigma(14182439040) = 70912195200 = 5 \times 14182439040$ OEIS A046060
Perfect cube	A <i>perfect cube</i> is a number equal to k^3 for some integer k .	125: $125 = 5^3$ OEIS A000578
Perfect number	A <i>perfect number</i> is an integer equal to the sum of its proper divisors. $s(n) = n$, $\sigma(n) = 2n$.	28: Aliquot sum: $s(28) = 1 + 2 + 4 + 7 + 14 = 28$ OEIS A000396
Perfect power	A <i>perfect power</i> is a number of the form m^k with integers $m > 1$, $k > 1$.	81: $81 = 3^4$, $m = 3$, $k = 4$ OEIS A001597
Perfect square	A <i>perfect square</i> is a number equal to k^2 for some integer k .	100: $100 = 10^2$ OEIS A000290
Perfect totient number	A <i>perfect totient number</i> is an integer that equals the sum of its iterated totients (the count of integers $\leq n$ that are coprime to n), including the final 1. $n = \varphi(n) + \varphi(\varphi(n)) + \varphi(\varphi(\varphi(n))) + \dots + 1$	9: $9 = 3^2$ The formula for Euler's totient is: $\varphi(n) = n \prod_{p n} \left(1 - \frac{1}{p}\right)$ where the product runs over the distinct prime factors p of n . $\varphi(9) = 9 \left(1 - \frac{1}{3}\right) = 9 \times \frac{2}{3} = 6$, $\varphi(6) = 2$, $\varphi(2) = 1$, Sum: $6 + 2 + 1 = 9$ OEIS A082897

Classifier	Description / Definition	Examples / OEIS reference
Polite number	A <i>polite number</i> is a positive integer that can be expressed as the sum of two or more consecutive positive integers. $n = a + (a + 1) + (a + 2) + \dots + (a + k - 1)$ for some integer $a \geq 1$ and $k \geq 2$. All integers except the powers of 2 are polite.	9: $9 = 4 + 5$ and also: $9 = 2 + 3 + 4, \Rightarrow 9$ is a polite number. OEIS A138591
Powerful number (Squareful number)	A <i>powerful number</i> is a positive integer n in which every prime factor appears with an exponent of at least 2. If $n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$ is the prime factorization of n, then n is powerful if $a_i \geq 2$ for every i.	72: Prime factors: $2^3 \times 3^2$, All exponents are $\geq 2 \Rightarrow 72$ is a powerful number. OEIS A001694
Practical number	A practical number is a positive integer n such that every smaller positive integer (from 1 up to $n - 1$) can be expressed as a sum of distinct divisors of n. In other words, using only the divisors of n (each at most once), you can form <i>any</i> smaller number by addition.	12: Divisors : 1, 2, 3, 4, 6, 12. All integers 1-11 can be expressed as sums of distinct divisors $\Rightarrow 12$ is a practical number. OEIS A005153
Primary pseudoperfect number	A positive integer $n > 1$ is called a <i>primary pseudoperfect number</i> if $\sum_{p n} \frac{1}{p} + \frac{1}{n} = 1$ where the sum is taken over all distinct prime divisors p of n.	42: Prime divisors: $2 \times 3 \times 7$: $\frac{1}{2} + \frac{1}{3} + \frac{1}{7} + \frac{1}{42} = \frac{21+14+6+1}{42} = \frac{42}{42} = 1 \Rightarrow 42$ is a primary pseudoperfect number. OEIS A054377
Primitive weird number	A <i>primitive weird number</i> is a weird number that has no weird proper divisors — that is, it is <i>weird</i> but not a multiple of any smaller weird number.	70: Sum of proper divisors: $s(70) = 1 + 2 + 5 + 7 + 10 + 14 + 35 = 74 > 70 \rightarrow$ abundant. No subset of $\{1, 2, 5, 7, 10, 14, 35\}$ adds up to 70 $\rightarrow$ not semi-perfect, So 70 is a weird number, and since there are no smaller weird numbers dividing 70, it is also a primitive weird number. OEIS A006037
q-aspiring number	Under the reduced aliquot iteration $q(n) = s(n) - 1$, the sequence reaches a perfect number.	8: Nontrivial proper divisors 2,4. $q(8) = s(8) - 1 = (1 + 2 + 4) - 1 = 6$ (equivalently $2 + 4 = 6$). q-sequence: $8 \rightarrow 6 \rightarrow 5 \rightarrow 0$, the sequence hits perfect 6 at step 1 $\Rightarrow 8$ is q-aspiring. OEIS N/A

Classifier	Description / Definition	Examples / OEIS reference
q-sociable number (Reduced sociable number)	Under the reduced aliquot iteration $q(n) = s(n) - 1$, cycles back to the starting number, with a minimum cycle length ≥ 3 .	1215571544 : Sequence $q(n)$: 1215571544 $\rightarrow$ 1270824975 $\rightarrow$ 1467511664 $\rightarrow$ 1530808335 $\rightarrow$ 1579407344 $\rightarrow$ 1638031815 $\rightarrow$ 1727239544 $\rightarrow$ 1512587175 $\rightarrow$ 1215571544 (cycle length = 8) OEIS A309227
q-socially aspiring number	Under the reduced aliquot iteration $q(n) = s(n) - 1$, eventually reaches a finite q-cycle of length ≥ 2 that does not contain n itself.	92 : Sum of proper divisors: $q(92) = 2 + 4 + 23 + 46 = 75$, $\Rightarrow$ $q(75) = 48$, $q(48) = 75$ So the q-sequence is: 92 $\rightarrow$ 75 $\rightarrow$ 48 $\rightarrow$ 75 $\rightarrow$ 48 $\rightarrow$... The pair {75,48} is a q-cycle of length 2. 92 is not in the cycle, and it enters it at step 1. $\Rightarrow$ 92 is a q-socially aspiring number. OEIS N/A
Quadriperfect number	A <i>quadriperfect number</i> (also called a 4-multiperfect number) is an integer whose sum of divisors equals four times the number itself. $\Sigma(n) = 4n$	30240 : 96 divisors summing to $\sigma(30240) = 120960 = 4 \times 30240$ $\rightarrow$ 30240 is a quadriperfect number. OEIS A027687
Quasi-sociable number	Under s-iteration, the aliquot sequence enters a sociable cycle (len $\geq$ 2) not containing n.	562 : Sum of proper divisors: $s(562) = 284$, the s-sequence is: 562 $\rightarrow$ 284 $\rightarrow$ 220 $\rightarrow$ 284 $\rightarrow$... This enters a sociable cycle of $k = 2$ at step 1, and 562 is not in the cycle $\Rightarrow$ it's quasi-sociable. OEIS A309227
Semiperfect number (Pseudoperfect number)	A <i>semiperfect number</i> is a positive integer that equals the sum of some (or all) of its proper divisors.	20 : Proper divisors: 1,2,4,5,10. check subsets sums to 20: 10 + 5 + 4 + 1 = 20 $\Rightarrow$ 20 is a semiperfect number. OEIS A005835
Sociable number	Aliquot sequence forms a cycle that returns to the starting number.	12496 : Aliquot sequence: $s(12496) = 12496 \rightarrow 14288 \rightarrow$ 15472 $\rightarrow$ 14536 $\rightarrow$ 14264 $\rightarrow$ 12496 OEIS A003416
Socially aspiring number	Aliquot sequence reaches an amicable or sociable cycle without including n in that cycle.	562 : Aliquot sequence: $s(562) = 284$, $s(284) = 220$, $s(220) = 284$. The pair 220 $\leftrightarrow$ 284 is an amicable (sociable-2) cycle. 562 reaches that 2-cycle at step 1, but is not itself in it $\Rightarrow$ 562 is a socially aspiring number (it enters a sociable 2-cycle at step 1). OEIS A121508

Classifier	Description / Definition	Examples / OEIS reference
Solitary number	Shares its abundancy ratio with no other integer; sufficient condition: $\gcd(n, \sigma(n)) = 1$.	3: Sum of divisors: $\sigma(3) = 1 + 3 = 4$ $\frac{\sigma(3)}{3} = \frac{4}{3}$ $\gcd(3, 4) = 1$ which satisfies the condition $\Rightarrow 3$ is a solitary number. OEIS N/A
Sphenic number	A <i>sphenic number</i> is a positive integer that is the product of three distinct prime numbers, each appearing exactly once in the factorization.	30: Prime factors: $30 = 2 \times 3 \times 5$. All three factors are prime and distinct $\Rightarrow 30$ is a sphenic number. OEIS A007304
Squarefree number	A <i>square-free number</i> is a positive integer where no prime factor appears with an exponent > 1 .	42: $42 = 2^1 \times 3^1 \times 7^1$ Each prime factor appears only once $\Rightarrow 42$ is square-free. OEIS A005117
Sublime number	Number with a perfect number of divisors, and their sum is also perfect. Only two sublime numbers are known.	12: Sum of divisors: $\sigma(12) = 1 + 2 + 3 + 4 + 6 + 12 = 28$, number of divisors = 6. Both 6 and 28 are perfect numbers $\Rightarrow 12$ is sublime. OEIS A081357
Superabundant number	A number n is <i>superabundant</i> if $\frac{\sigma(n)}{n} > \frac{\sigma(m)}{m}$ for every $m < n$.	12: Sum of divisors: $\sigma(12) = 1 + 2 + 3 + 4 + 6 + 12 = 28$ Ratio: $\frac{\sigma(12)}{12} = \frac{28}{12} = 2.333 \dots$ All other smaller numbers do not have a higher $\sigma(12n)/12$ ratio $\Rightarrow 12$ is superabundant. OEIS A004394
Superperfect number	A superperfect number is a positive integer n for which the sum of the divisors of the sum of its divisors equals twice the number itself. $\sigma(\sigma(n)) = 2n$.	16: Sum of divisors: $\sigma(16) = 1 + 2 + 4 + 8 + 16 = 31$ $\sigma(31) = 1 + 31 = 32$ $\sigma(\sigma(16)) = 32 = 2 \times 16 \Rightarrow 16$ is superperfect. OEIS A019279
Triperfect number	A <i>triprfect number</i> (also called a 3-perfect number) is an integer whose sum of divisors equals three times the number itself. $\Sigma(n) = 3n$	120: Sum of divisors: $\sigma(120) = 1 + 2 + 3 + 4 + 5 + 6 + 8 + 10 + 12 + 15 + 20 + 24 + 30 + 40 + 60 + 120 = 360$ $360 / 120 = 3$ OEIS A005820
Unitary perfect number	A <i>unitary perfect number</i> is a positive integer n for which the sum of its unitary divisors equals twice the number itself: $\sigma^*(n) = 2n$	60: Prime factors: $2^2 \times 3 \times 5$ $\sigma^*(60) = (1 + 2^2)(1 + 3)(1 + 5) = 5 \times 4 \times 6 = 120$. $120 / 60 = 2 \Rightarrow 60$ is a unitary perfect number. OEIS A002827

Classifier	Description / Definition	Examples / OEIS reference
Untouchable number	An <i>untouchable number</i> is a positive integer that cannot be written as the sum of the proper divisors $s(m)$ for any $m > 0$. In other words, no integer has this number as its <i>aliquot sum</i> .	5: For all primes p $s(p) = 1$, leaving to check: $s(1) = 0$, $s(4) = 3$, $s(6) = 6$, $s(8) = 7$. After $m = 8$, $s(m)$ already exceeds 5 for all n so no new n can give 5. (the only known odd one). OEIS A005114
Weird number	A <i>Weird number</i> is an abundant number that is not semiperfect, that is, the sum of its proper divisors exceeds the number itself, but no subset of those divisors adds up exactly to the number.	70: Sum of proper divisors: $s(70) = 1 + 2 + 5 + 7 + 10 + 14 + 35 = 74 > 70$. However, there is no subset of 1,2,5,7,10,14,35 that sums exactly to 70 $\Rightarrow$ 70 is weird, the smallest known. OEIS A006037
Zumkeller number	A <i>Zumkeller number</i> is a positive even integer whose divisors can be partitioned into two disjoint subsets having equal sums.	12: Sum of divisors: $\sigma(12) = 1 + 2 + 3 + 4 + 6 + 12 = 28$, $28 / 2 = 14$. We can partition the divisors into: $\{1,3,4,6\}$ and $\{2,12\}$. Since $1 + 3 + 4 + 6 = 14 = 2 + 12$. $\Rightarrow$ 12 is a Zumkeller number. OEIS A083207

Combinatorial and Geometric Numbers

Numbers that arise from discrete counting principles — representing partitions, permutations, arrangements lattice paths or geometric divisions.

Note: Some figurate numbers (like Tetrahedral or Triangular) can also be expressed combinatorially via binomial coefficients. In this manual, they are grouped with Polygonal and Figurate Numbers for their geometric interpretation.

Classifier	Definition / Description	Examples / OEIS
Bell number	A <i>Bell number</i> counts the number of ways a set with n elements can be partitioned into non-empty, disjoint subset. $B(n) = \sum_{k=0}^n S(n, k)$ where $S(n, k)$ is a Stirling number of the second kind (the number of ways to divide n objects into k non-empty groups).	3: The set $\{a, b, c\}$ can be partitioned in 5 ways: 1. $\{\{a, b, c\}\}$ 2. $\{\{a, b\}, \{c\}\}$ 3. $\{\{a, c\}, \{b\}\}$ 4. $\{\{b, c\}, \{a\}\}$ 5. $\{\{a\}, \{b\}, \{c\}\}$ Hence $B(3) = 5$. OEIS A000110
Cake number (Maximum piece number)	A <i>cake number</i> gives the maximum number of pieces into which a solid cake (or any 3-dimensional object) can be divided using n straight planar cuts. Each new cut can intersect all previous cuts, thereby maximizing the number of separate regions produced.	26: $= C_n$ Multiply both sides by 6: $6C_n = n^3 + 5n + 6$ Rearrange into a cubic in n : $n^3 + 5n + (6 - 6C_n) = 0$ Let $P = 5$, $Q = 6 - 6C_n =$

Classifier	Definition / Description	Examples / OEIS
	Mathematically, the sequence counts regions of space cut by n planes in general position. The n-th cake number C_n is given by: $C_n = \frac{n^3 + 5n + 6}{6}$ and satisfies the recurrence $C_n = C_{n-1} + \binom{n}{2} + 1$ Reverse function according to Cardano's formula for the real root of $x^3 + Px + Q = 0$: $x = \sqrt[3]{-\frac{Q}{2} + \sqrt{\left(\frac{Q}{2}\right)^2 + \left(\frac{P}{3}\right)^3}} + \sqrt[3]{-\frac{Q}{2} - \sqrt{\left(\frac{Q}{2}\right)^2 + \left(\frac{P}{3}\right)^3}}$	$6 - 6 \times 26 = -150$. $n = \sqrt[3]{-\frac{-150}{2} + \sqrt{\left(\frac{-150}{2}\right)^2 + \left(\frac{5}{3}\right)^3}} + \sqrt[3]{-\frac{-150}{2} - \sqrt{\left(\frac{-150}{2}\right)^2 + \left(\frac{5}{3}\right)^3}}$ $n = \sqrt[3]{75 + \sqrt{5629.63 \dots}} + \sqrt[3]{75 - \sqrt{5629.63 \dots}} = 5$ $\Rightarrow$ 26 pieces correspond to 5 straight cuts. OEIS A000125
Catalan number	The <i>Catalan numbers</i> form a famous sequence in combinatorics that counts many kinds of structured objects. They appear in problems involving balanced parentheses, binary trees, polygon triangulations, and lattice paths. The n-th Catalan number is given by the formula: $C_n = \frac{1}{n+1} \binom{2n}{n} = \frac{(2n)!}{(n+1)! n!}$	3: $C_3 = \frac{1}{4} \binom{6}{3} = \frac{1}{4} \times 20 = 5$. Interpretation: There are 5 distinct ways to correctly match 3 pairs of parentheses: $((()))$, $((()))$, $(())()$, $()(())$, $()()()$ or equivalently, 5 distinct binary trees with 3 internal nodes. OEIS A000108
Fubini number (Ordered Bell number)	A <i>Fubini number</i> counts the number of weak orders or total preorders on an n-element set. Equivalently, it counts the number of ways to partition a set of n labeled elements into non-empty subsets where each block is ordered, and the blocks themselves are ordered. $F(n) = \sum_{k=1}^n k! S(n, k)$	13: verify from the formula: $F(n) = \sum_{k=0}^n k! S(n, k)$ Take $n=3$. Using Stirling numbers of the second kind $S(3, k)$: $S(3,1) = 1$ $S(3,2) = 3$ $S(3,3) = 1$ $F(3) = 1! \times 1 + 2! \times 3 + 3! \times 1 = 1 + 6 + 6 = 13 \Rightarrow 13$ is the third Fubini number. OEIS A000670
Lah number	A <i>Lah number</i>, written $L(n, k)$, counts the number of ways to partition n labeled elements into k non-empty linearly ordered subsets (i.e. ordered <i>lists</i>, not just sets). Unlike Stirling numbers (which count unordered subsets), Lah numbers take ordering within each subset into account. $L(n, k) = \frac{n!}{k!} \binom{n-1}{k-1}$	141120: Take $n = 8$ For $k = 2$: $L(8,2) = \frac{8!}{2!} \binom{7}{1} = \frac{40320}{2} \times 7 = 20160 \times 7 = 141120$ For $k = 3$: $L(8,3) = \frac{8!}{3!} \binom{7}{2} = \frac{40320}{6} \times 21 = 6720 \times 21 = 141120$ $\Rightarrow 141120 = L(8,2) = L(8,3)$

Classifier	Definition / Description	Examples / OEIS																		
	for integers $n, k \geq 1$.	OEIS A105278																		
Motzkin number	A <i>Motzkin number</i> $M(n)$ counts the number of lattice paths of length n that start at $(0, 0)$, end at $(n, 0)$, and never go below the x-axis, using only the steps Up $(1, 1)$, Flat $(1, 0)$, Down $(1, -1)$. Equivalently, it counts the number of ways to draw non-intersecting chords connecting n labeled points on a circle, or the number of non-crossing matchings in combinatorics. The numbers satisfy the recurrence $M(0) = 1, M(1) = 1, M(n) = M(n-1) + \sum_{k=0}^{n-2} M(k) M(n-2-k) (n \geq 2)$, and can also be expressed as $M(n) = \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \binom{n}{2k} C_k,$ where C_k is the k-th Catalan number.	4: $M(0) = 1, M(1) = 1, M(2) = 2, M(3) = 4, M(4) = 9, M(5) = 21, \dots$ Illustration ($n = 3$): All paths of length 3 from $(0, 0)$ to $(3, 0)$ staying above the x-axis: 1. F F F (flat, flat, flat) 2. U D F (up, down, flat) 3. U F D (up, flat, down) 4. F U D (flat, up, down) Hence $M(3) = 4 \Rightarrow 4$ is a Motzkin number. OEIS A001006																		
Partition number	The <i>partition number</i> $p(n)$ counts the number of distinct ways a positive integer n can be written as a sum of positive integers, disregarding order of the terms. Formally, each decomposition $n = a_1 + a_2 + \dots + a_k (a_i \geq 1)$ is a <i>partition</i> of n, where the sequence of summands is unordered (so $4 = 3 + 1 = 1 + 3$ counts as one partition).	7: <table border="1"> <thead> <tr> <th>N</th><th>partitions</th><th>p(n)</th></tr> </thead> <tbody> <tr> <td>1</td><td>1</td><td>1</td></tr> <tr> <td>2</td><td>2; 1+1</td><td>2</td></tr> <tr> <td>3</td><td>3; 2+1; 1+1+1</td><td>3</td></tr> <tr> <td>4</td><td>4; 3+1; 2+2; 2+1+1; 1+1+1+1</td><td>5</td></tr> <tr> <td>5</td><td>5; 4+1; 3+2; 3+1+1; 2+2+1; 2+1+1+1; 1+1+1+1+1</td><td>7</td></tr> </tbody> </table> $\Rightarrow$ partition number $p(5) = 7$ OEIS A000041	N	partitions	p(n)	1	1	1	2	2; 1+1	2	3	3; 2+1; 1+1+1	3	4	4; 3+1; 2+2; 2+1+1; 1+1+1+1	5	5	5; 4+1; 3+2; 3+1+1; 2+2+1; 2+1+1+1; 1+1+1+1+1	7
N	partitions	p(n)																		
1	1	1																		
2	2; 1+1	2																		
3	3; 2+1; 1+1+1	3																		
4	4; 3+1; 2+2; 2+1+1; 1+1+1+1	5																		
5	5; 4+1; 3+2; 3+1+1; 2+2+1; 2+1+1+1; 1+1+1+1+1	7																		
Ramsey number (2-colour, exact)	A 2-colour <i>Ramsey number</i> $R(r, s)$ is the smallest integer N such that every red/blue colouring of the edges of the complete graph K_N contains either a red K_r or a blue K_s. Because the definition is symmetric, $R(r, s) = R(s, r)$. Only a few values are known exactly. Small exact values include $R(3,3) = 6, R(3,4) = 9, R(3,5) = 14, R(4,4) = 18, \dots$ For diagonal Ramsey numbers $R(k, k)$, exact values are known only for $k \leq 4$. For $k \geq 5$, a conjectured estimate is used for (k, k). $a(0) = 0, a(1) = 1, a(2) = 2,$ $a(k) = \left\lceil \left(\frac{3}{2} \right)^{k-3} k(k-1) \right\rceil (k \geq 3)$	6: For example, $R(3,3)=6$: in any red/blue coloring of the edges of a complete graph on six vertices, there must be a monochromatic triangle (a 3-clique all in one color), whereas for five vertices it is still possible to avoid one. $N=6 \Rightarrow R(3,3)=6$ Any 2-coloring of the graph K_6 will contain either a red triangle K_3 or a blue triangle K_3. $\Rightarrow 6 = \text{Ramsey number } R(3,3)$ OEIS A059442 (exact values) OEIS A120414 (conjectured diagonal values)																		

Classifier	Definition / Description	Examples / OEIS
Rooted tree number	A rooted tree is a connected, acyclic graph with one distinguished vertex called the root. A rooted tree number gives the number of non-isomorphic rooted (unlabeled) trees with n vertices. Trees that differ only by renaming their vertices are counted as the same tree.	20: 20 is a rooted tree number belonging to 6 vertices. OEIS: A000081
Schur number (v1) exact	The Schur number $v_1 S_1(k)$ is the largest integer n such that the entire interval $\{1, 2, \dots, n\}$ can be colored with k colors with the property that there is no monochromatic solution to the equation $x + y = z$. Equivalently: $S_1(k)$ is the maximum length of an initial interval that avoids any single-colored triple (x, y, z) with $x + y = z$. Beyond $n = S_1(k)$, any k -coloring of $1 \dots n$ necessarily contains such a triple. These numbers are smaller than the classical Schur numbers (v2), because v1 requires coloring the full interval rather than arbitrary subsets.	5: For $k = 2$: $S_1(2) = 5$. - There exists a 2-colouring of $\{1, 2, 3, 4, 5\}$ avoiding any single-color solution to $x + y = z$. - But for $\{1, 2, 3, 4, 5, 6\}$, every 2-colouring gives a monochromatic sum triple (e.g. $1+5=6$ or $2+4=6$ in one color). $\Rightarrow 5$ is maximal, so $5 = S_1(2)$ OEIS A030126
Stirling number (unsigned, 1st kind)	The <i>Stirling numbers</i> of the first kind, written as $c(n, k)$ or $\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]$, count how many permutations of n distinct elements have exactly k cycles in their cycle decomposition, in other words: $\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]$ = number of ways to arrange n objects into k disjoint cycles. These numbers arise naturally when expressing falling factorials in terms of ordinary powers, or in studying the structure of permutations.	11: For $n=4$: The permutations of 4 elements can be partitioned by number of cycles: $c(4, 1)=6$ permutations with 1 cycle (the 6 full 4-cycles) $c(4, 2)=11$ permutations with 2 cycles $c(4, 3)=6$ permutations with 3 cycles $c(4, 4)=1$ permutation with 4 cycles (the identity) Thus the 4th row of the unsigned triangle is: $4! = 24 = c(4, 1) + c(4, 2) + c(4, 3) + c(4, 4) = 6 + 11 + 6 + 1 \Rightarrow 11 =$ Stirling number 1 st kind $c(4, 2)$ OEIS A132393
Stirling number (2nd kind)	The <i>Stirling numbers of the second kind</i> , written as $S(n, k)$ or $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$, count the number of ways to partition a set of n distinct elements into k non-empty, unlabeled subsets (called <i>blocks</i>), in other words:	7: For $n=4$ and $k=2$ $S(4, 2)=7$. There are 7 ways to partition the set $\{1, 2, 3, 4\}$ into 2 non-empty subsets: $\{1, 2, 3\}, \{4\}$ $\{1, 2, 4\}, \{3\}$ $\{1, 3, 4\}, \{2\}$ $\{2, 3, 4\}, \{1\}$ $\{1, 2\}, \{3, 4\}$

Classifier	Definition / Description	Examples / OEIS
	$\{n\}_k$ = number of ways to divide n objects into k groups, ignoring the order of the groups.	6. $\{1,3\}, \{2,4\}$ 7. $\{1,4\}, \{2,3\}$ $\Rightarrow 7$ = Stirling number 2 nd kind $S(4,2)$ OEIS A008277
Van der Waerden number (exact)	A <i>Van der Waerden</i> number $W(r, k)$ is the smallest positive integer N such that every r -coloring of the integers $\{1, 2, \dots, N\}$ contains a monochromatic arithmetic progression of length k . In other words: no matter how you color the first N integers with r colors, you cannot avoid producing a progression $a, a + d, a + 2d, \dots, a + (k - 1)d$ whose elements all share the same color.	9: $W(2,3) = 9$ Any coloring of $\{1, \dots, 9\}$ with two colors will contain a monochromatic arithmetic progression of three terms (e.g., 1–5–9 or 2–5–8). For $n = 8$ it can still be avoided, but for $n = 9$ the property is unavoidable $\Rightarrow 9$ is a Van der Waerden number. OEIS A005346

Conjectures and Equation-based Numbers

Numbers appearing in or related to classical conjectures and Diophantine equations.

Classifier	Definition / Description	Examples / OEIS
Egyptian m/n	Searches m/n as a sum of k unit fractions. $\frac{4}{n} = \frac{1}{x} + \frac{1}{y}$ Use <code>egyptian_m_value</code> and <code>egyptian_k_value</code> in profile, default: $m = 4, k = 2$	11: $\frac{4}{11} = \frac{1}{3} + \frac{1}{33}$ (for $m = 1, n = 11, k = 4$) OEIS A192881
Erdős–Straus	Tests whether $\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ has a solution in positive distinct integers. Conjectured true for all $n \geq 2$.	5: $\frac{4}{5} = \frac{1}{2} + \frac{1}{4} + \frac{1}{20}$ OEIS A073101
Goldbach conjecture	Every even integer > 2 can be written as the sum of two primes.	8: $3 + 5 = 8$ OEIS A002372
Legendre prime interval	<i>Legendre's conjecture</i> states that for every $n > 0$ there exists at least one prime p such that $n^2 < p < (n + 1)^2$.	5: $n = 5, n^2 = 25, (n + 1)^2 = 36$, in that range we have two primes: 29 and 31. OEIS A014085
Lemoine's (Levy's)	Every odd integer > 5 can be written as $p + 2q$ with primes p, q .	53: $47 + 2 \times 3 = 53$ (has 5 different solutions in total.) OEIS A194828
Sierpiński	<i>Sierpiński's conjecture</i> asserts that for every integer $n \geq 2$, the Diophantine equation $5/n = 1/x + 1/y + 1/z + 1/w$ has a solution in positive integers.	9: $\frac{5}{9} = \frac{1}{3} + \frac{1}{9} + \frac{1}{10} + \frac{1}{90}$ OEIS N/A

Weak Goldbach (ternary)	Every odd integer ≥ 7 can be expressed as the sum of three primes.	91: $3 + 5 + 83 = 91$ (91 has 83 different solutions in total.) OEIS A068307
Znám chain	A proper Znám set $\{a_1, \dots, a_k\}$ satisfies $\sum_{i=1}^k \frac{1}{a_i} = 1 - \frac{1}{(a_1 a_2 \dots a_k)}$ The Znám chain of length k finds such a set. k range: proper version $k \in [5, 10]$; improper version: $k \in [2, 10]$ and allows the largest element to equal the product of the others plus 1 (i.e., $a_{\max} = \prod_{i \neq \max} a_i + 1$).	6: $S = \frac{1}{2} + \frac{1}{3} + \frac{1}{7} + \frac{1}{47} + \frac{1}{403} + \frac{1}{19403} = 1 - \frac{1}{P}$ $P = 2 \times 3 \times 7 \times 47 \times 403 \times 19403 = 15435513366$ OEIS A075441

Digit-based Numbers

Numbers defined by patterns in their digits (usually base 10).

Classifier	Definition / Description	Examples / OEIS
Automorphic number	An <i>Automorphic number</i> is an integer whose square ends in the same digits as the number itself. n^2 ends with n .	76: $76^2 = 5776$ OEIS A003226
Brazilian number	A <i>Brazilian number</i> can be represented as a repdigit (repeating copies of the same digit) in some base b with $2 \leq b \leq n - 2$.	72: In base 11, 72 is a repdigit of length 2 with digit 6: $66 = 6 \times R_2(11)$, where $R_2(11) = \frac{11^2 - 1}{11 - 1} = \frac{120}{10} = 12$ $n = 6 \times 12 = 72$ OEIS A125134
Disarium number	A <i>Disarium number</i> is a number that is equal to the sum of its digits each raised to the power of their respective positions (counting from left to right, starting at 1). It has been proven that no term can have more than 22 digits, and a complete search up to 10^{22} shows that there are only 20 Disarium numbers with the last being: 12157692622039623539.	135: $1^1 + 3^2 + 5^3 = 1 + 9 + 125 = 135$ $\Rightarrow 135$ is a Disarium number. OEIS A032799
Dudeney number	A <i>Dudeney number</i> (sometimes spelled Dudeney number) has the cube of the sum of digits equal to the number itself. $((\text{sum of digits})^3 = n)$ Only six such numbers are known.	512: $5 + 1 + 2 = 8$, $8^3 = 512$ $\Rightarrow 512$ is a Dudeney number. OEIS A061209
Evil number	An <i>Evil number</i> is a non-negative integer that has an even number of 1-bits in its binary (base-2) representation.	9: $9_{10} = 1001_2$ The binary form has two 1's, which is an even number. $\Rightarrow 9$ is an Evil number. OEIS A001969

Classifier	Definition / Description	Examples / OEIS
Factorion	A <i>Factorion</i> is a number that is equal to the sum of the factorials of its digits.	145: $1! + 4! + 5! = 1 + 24 + 120 = 145 \Rightarrow 145$ is a Factorion. OEIS A014080
Fibonacci-concatenation	A <i>Fibonacci concatenation</i> number is a number formed by writing several consecutive Fibonacci numbers next to each other (seed 1,1 or 0,1).	11235813: Fibonacci numbers, (seed 1,1) 1, 1, 2, 3, 5, 8, 13, ... Concatenated digits: 11235813 $\Rightarrow$ is a Fibonacci-concatenation. OEIS A019523
Grafting number	<i>Grafting numbers</i> are integers that, when expressed in base 10, will appear in their own square root before or directly after the decimal point (ignoring leading 0's and including trailing 0's).	765: $\sqrt{765} \approx 27.65863$ the square root contains the digits 7. 65 around the decimal point $\Rightarrow 765$ is a grafting number. OEIS A232087
Happy number	A <i>Happy number</i> is a number that eventually reaches 1 when you repeatedly replace it by the sum of the squares of its digits.	19: $1^2 + 9^2 = 82, 8^2 + 2^2 = 68, 6^2 + 8^2 = 100, 1^2 + 0^2 + 0^2 = 1$ $\Rightarrow 19$ is a Happy number. OEIS A007770
Harshad number (Niven number)	A <i>Harshad number</i> is an integer that is divisible by the sum of its digits in a given base (usually base 10).	18: Sum of digits: $1 + 8 = 9, 18 \div 9 = 2 \Rightarrow 18$ is a Harshad number. OEIS A005349
Hoax number	A <i>Hoax number</i> is a composite number whose digit sum equals the sum of the digits of its distinct prime factors.	319: Prime factors: 11×29 Sum of factor digits: $1 + 1 + 2 + 9 = 13$. Sum of digits of 319: $3 + 1 + 9 = 13 \Rightarrow 319$ is a Hoax number. OEIS A019506
Kaprekar number	A <i>Kaprekar number</i> is a non-negative integer whose square can be split into two parts that, when added together, give back the original number.	45: $45^2 = 2025$, split into 20 and 25: $20 + 25 = 45 \Rightarrow 45$ is a Kaprekar number. OEIS A006886
Lychrel number	A <i>Lychrel number</i> is a natural number that is suspected never to form a palindrome through the reverse-and-add process. NumClass sets the limit to 1000 iterations. After reaching the limit the number is considered to be a Lychrel number candidate.	196: $196 + 691 = 887, 887 + 788 = 1675, 1675 + 5761 = 7436, 7436 + 6347 = 13783$, etc. No palindrome found after 1000 iterations $\Rightarrow 196$ is a Lychrel number candidate. OEIS A023108
Moran number	A <i>Moran number</i> is a positive integer that is divisible by the sum of its digits, and when divided by that sum, the quotient is a prime number.	133: Sum of digits: $1 + 3 + 3 = 7$. $133 \div 7 = 19$, 19 is prime $\Rightarrow 133$ is a Moran number. OEIS A001101
Narcissistic number Armstrong number	A Narcissistic number is a number that is equal to the sum of its own digits each raised to the power of the number of digits.	153: has 3 digits: $1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153 \Rightarrow 153$ is a Narcissistic number. OEIS A005188

Classifier	Definition / Description	Examples / OEIS
Odious number	An <i>Odious number</i> is a non-negative integer that has an odd number of 1-bits in its binary (base-2) representation.	11: $11_{10} = 1011_2$ The binary form has three 1's, which is an odd number. $\Rightarrow$ 11 is an Odious number. OEIS A000069
Palindrome	A <i>Palindrome</i> reads identically forward and backward in decimal notation.	13631: Reads the same forward and backward $\Rightarrow$ 13631 is a Palindrome. OEIS A002113
Palindromic Harshad number (intersection)	A <i>Palindromic Harshad number</i> is a Harshad number that is also a palindrome in decimal form.	111: Sum of digits: $1 + 1 + 1 = 3$ $111 \div 3 = 37 \Rightarrow$ 111 is a Harshad number and palindromic. OEIS A082232
Pernicious number	A <i>pernicious number</i> is a positive integer that has a prime number of 1s (a "prime Hamming weight") in its binary representation. In other words, when you convert the number to base 2, the sum of the digits is a prime number.	14: $14_{10} = 1110_2$ The sum is $1+1+1 = 3$, 3 is a prime number $\Rightarrow$ 14 is a Pernicious number. OEIS A052294
Self number (Devlali number) (Colombian number)	A <i>Self number</i> is a positive integer that cannot be generated by any other number through the function $f(n) = n + \text{sum of digits of } n$. In other words, there is no integer m such that $m + \text{sum of digits of } m = n$. If such an m does not exist, then it "creates itself," because no other number produces it.	20: Try to find m such that $m + \text{sum of digits of } m = 20$. • $15 + (1 + 5) = 21$ • $14 + (1 + 4) = 19$ • $13 + (1 + 3) = 17$ • $12 + (1 + 2) = 15$ None produce 20 $\Rightarrow$ 20 is a Self number. OEIS A003052
Self-descriptive number (Self-documenting number)	A <i>Self-descriptive number</i> is an integer that describes itself — each digit in the number tells how many times a particular digit appears in the whole number. In base 10, the digit in position i (counting from 0 on the left) represents how many times the digit i occurs in the entire number.	6210001000: Position 0 1 2 3 4 5 6 7 8 9 Digit 6 2 1 0 0 0 1 0 0 0 Check: There are six 0's, two 1's, one 2, zero 3's, zero 4's, zero 5's, one 6, zero 7's, 8's and 9's . All conditions hold $\Rightarrow$ 6210001000 is a Self-descriptive number. OEIS A108551
Smith number (Joke number)	A <i>Smith number</i> is a composite number whose sum of digits equals the sum of the digits of its prime factors (counted with multiplicity). In other words, when you factorize the number into primes, and add up all the digits appearing in those factors, the total matches the sum of the digits of the original number.	666: Prime factors: $2 \times 3 \times 3 \times 37$, Sum of digits: $6 + 6 + 6 = 18$. Sum of digits of all prime factors (with multiplicity): $(2) + (3 + 3) + (3 + 7) = 2 + 3 + 3 + 3 + 7 = 18 \Rightarrow$ 666 is a Smith number. OEIS A006753

Classifier	Definition / Description	Examples / OEIS
Sum of 2 palindromes	Can be expressed as the sum of two palindromic numbers in base 10.	167: $66 + 101 = 167$ OEIS A035137
Sum of 3 palindromes	Can be expressed as the sum of three palindromic numbers in base 10. It is proved that every integer is a sum of three palindromes.	500: $77 + 151 + 272 = 500$ OEIS A261132

Diophantine Representations

Numbers expressible as sums of integer powers, forming classical Diophantine identities.

Classifier	Definition / Description	Examples / OEIS
Sum of 2 cubes	Can be written as $n = a^3 + b^3$ for integers a, b.	28: $1^3 + 3^3 = 1 + 27 = 28$ OEIS A003325
Sum of 2 squares	Can be written as $n = a^2 + b^2$ for integers a, b.	13: $2^2 + 3^2 = 9 + 4 = 13$ OEIS A001481
Sum of 3 cubes	Can be written as $n = a^3 + b^3 + c^3$ for integers a, b, c.	7: $(-105)^3 + 32^3 + 104^3 = 7$ OEIS A060464
Sum of 3 squares	Can be written as $n = a^2 + b^2 + c^2$ for integers a, b, c.	29: $2^2 + 3^2 + 4^2 = 4 + 9 + 16 = 29$ OEIS A000378

Dynamical Sequences

Numbers generated or studied through iterative or recursive rules.

Classifier	Definition / Description	Examples / OEIS
Collatz (3n + 1) sequence	Defined by the rule: $n \rightarrow n / 2$ if n is even, else $n \rightarrow 3n + 1$; conjectured that every positive integer eventually reaches 1.	6: sequence $6 \rightarrow 3 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$ 8 steps, 6 even, 2 odd, odd/even $1:3 \approx 0.333$ (25.0%), peak value 16. OEIS A006370
Generalized Collatz (5n + 1)	Defined by the rule: $n \rightarrow n / 2$ if n is even, else $n \rightarrow 5n + 1$	10: sequence $10 \rightarrow 5 \rightarrow 26 \rightarrow 13 \rightarrow 66 \rightarrow 33 \rightarrow 166 \rightarrow 83 \rightarrow 416 \rightarrow 208 \rightarrow 104 \rightarrow 52 \rightarrow 26$. Loops back to 26, peak value 416. OEIS A232711
Generalized Collatz (7n + 1)	Defined by the rule: $n \rightarrow n / 2$ if n is even, else $n \rightarrow 7n + 1$	10: sequence: $10 \rightarrow 5 \rightarrow 36 \rightarrow 18 \rightarrow 9 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$, peak value 64. OEIS A133421
Generalized Collatz (an + b)	Defined by the rule: $n \rightarrow n / 2$ if n is even, else $n \rightarrow an + b$, a and b can be specified in the profile settings. (Default: $3n+3$)	10: For default values $a=3$ and $b=3$: sequence: $10 \rightarrow 5 \rightarrow 18 \rightarrow 9 \rightarrow 30 \rightarrow 15 \rightarrow 48 \rightarrow 24 \rightarrow 12 \rightarrow 6 \rightarrow 3 \rightarrow 12$. Loops back to 12, peak value 48. OEIS N/A

Classifier	Definition / Description	Examples / OEIS
Ducci (4-digit) sequence	A Ducci sequence starts from the last four digits of the input number. These four digits form a tuple (a_1, a_2, a_3, a_4), and each step produces a new tuple of the absolute differences of adjacent entries (with wrap-around): $(a_1, a_2, a_3, a_4) \rightarrow (a_1 - a_2 , a_2 - a_3 , a_3 - a_4 , a_4 - a_1)$. A number is Ducci-terminating if the sequence eventually reaches $(0, 0, 0, 0)$.	91234: last for digits: 1234 $(1, 2, 3, 4) \rightarrow (1, 1, 1, 3)$ $(1, 1, 1, 3) \rightarrow (0, 0, 2, 2)$ $(0, 0, 2, 2) \rightarrow (0, 2, 0, 2)$ $(0, 2, 0, 2) \rightarrow (2, 2, 2, 2)$ $(2, 2, 2, 2) \rightarrow (0, 0, 0, 0)$ The sequence reaches all zeros after five steps, so 1234 is Ducci-terminating. OEIS N/A
Fibonacci mod n (Pisano period)	The <i>Pisano period</i> $\pi(n)$ is the length of the repeating cycle in the Fibonacci sequence taken modulo n. Formally, if $F_0 = 0$, $F_1 = 1$, $F_k = F_{k-1} + F_{k-2} \pmod{n}$, then the Pisano period $\pi(n)$ is the smallest positive integer m such that $F_m \equiv 0 \pmod{n}$ and $F_{m+1} \equiv 1 \pmod{n}$. The sequence of Fibonacci numbers modulo n always eventually repeats, and the period depends on n.	11: Fibonacci sequence mod 11: residues: $[0, 1, 1, 2, 3, 5, 8, 2, 10, 1, 0] \rightarrow$ repeat every 10 times, so the Pisano period $\pi(11) = 10$ OEIS A001175
Happy number sequence	Starting from any positive integer n, replace the number by the sum of the squares of its decimal digits. Repeat this process iteratively: $n \rightarrow \sum (\text{digits}(n))^2$. If this sequence eventually reaches 1, the original number is called a happy number.	19: $1^2 + 9^2 = 82$ $8^2 + 2^2 = 68$ $6^2 + 8^2 = 100$ $1^2 + 0^2 + 0^2 = 1 \Rightarrow 19$ is a happy number sequence, if not it's an unhappy number sequence. OEIS A007770
Kaprekar routine	Start with a positive integer, arrange its digits in descending and ascending order to form two numbers, and subtract the smaller from the larger. Repeat this process with each new result. For most numbers with a fixed number of digits, this process quickly reaches a fixed point or a short loop, known as a Kaprekar constant. - For 3-digit numbers $\rightarrow 495$ - For 4-digit numbers $\rightarrow 6174$ If all digits are identical (e.g., 1111), the process collapses to 0.	3524: $5432 - 2345 = 3087$ $8730 - 0378 = 8352$ $8532 - 2358 = 6174$ $\rightarrow$ The sequence $3524 \rightarrow 3087 \rightarrow 8352 \rightarrow 6174$ reaches the Kaprekar constant 6174, after which $7641 - 1467 = 6174$ repeats indefinitely. OEIS A099009

Classifier	Definition / Description	Examples / OEIS
Reverse-and-add sequence	Starting from any positive integer n, reverse its decimal digits and add the result to the original number: $n \rightarrow n + \text{reverse}(n)$ Repeat this process iteratively. If the sequence eventually becomes a palindrome (a number that reads the same forward and backward), n is said to reach a palindrome. Some numbers never appear to reach a palindrome — these are known as Lychrel candidates (for example, 196).	349: $349 + 943 = 1292$ $1292 + 2921 = 4213$ $4213 + 3124 = 7337$ $\rightarrow 7337$ is a palindrome after 3 steps. OEIS A033865

Fun Numbers

Numbers notable for cultural, historical, or humorous reasons rather than mathematical properties.

Classifier	Definition / Description	Examples / OEIS
Fun number	Recognized number of interest in computing, science fiction, or pop culture. The database currently contains 84 of such numbers.	42: The Answer to the Ultimate Question of Life, the Universe, and Everything (Douglas Adams, The Hitchhiker's Guide to the Galaxy) OEIS N/A

Note: All Fun Numbers are listed in file data/fun_numbers.tsv

Mathematical Curiosities

Numbers exhibiting aesthetic, linguistic, or self-referential properties.

Classifier	Definition / Description	Examples / OEIS
Additive sequence	Number whose digits can be split into parts forming an additive sequence where each term equals the sum of the two preceding ones.	437 : $4+3=7 \Rightarrow 437 \Rightarrow 437$ is an Additive sequence. OEIS A033627 (listing numbers that are NOT an Additive sequence.)
Binary-interpretable number	A decimal number that represents a valid binary numeral when its digits are read as bits.	1011 : Base 10 number 1011 can be interpreted as a number in base 2: $1+2+8=11$ in base 10. OEIS N/A
Block-power invariant	A number is block-power invariant if its decimal representation can be split into equal-length digit blocks, and the sum of those blocks raised to the same power gives back the original number. Leading zeroes inside blocks are allowed only if they occur naturally in the decimal split.	166500333 : split into 3-digit blocks: 166, 500, 333. $166^3 + 500^3 + 333^3 = 166500333$ $\Rightarrow 166500333$ is a block-power invariant number. OEIS: A056733 (for blocksize 3 and power 3 only)
Boring number	The first number with the least number of classifications in NumClass using the default profile.	1911 : The first number found having only 4 classifications. OEIS N/A
Cyclic permutation number	Integer whose cyclic rotations are successive multiples of the original number.	142857 : $142857 \times 1 = 142857$ $142857 \times 2 = 285714$ $142857 \times 3 = 428571$ $142857 \times 4 = 571428$ $142857 \times 5 = 714285$ $142857 \times 6 = 857142$ Each product is a cyclic rotation of the digits of 142857 $\Rightarrow 142857$ is a cyclic number OEIS A180340
Cyclops number	Has an odd number of digits with exactly one zero in the middle position.	11011 : Odd number and single zero in the middle. OEIS A134808
Digit-reversal constant	Appears as a fixed result in a digit-reversal subtraction procedure, reverse, subtract, reverse, add. This always converges to 1089.	532 : Reverse: 235, Subtract: $532-235=297$, Reverse: 792, Add: $297+792=1089$. OEIS N/A
Eban number	Lacks the letter “e” when its English name (short-scale) is spelled out. In English, every odd number contains an "e," so all eban numbers are even.	32 : (thirty-two). Does not contain the letter “e” $\Rightarrow 32$ is an eban number. OEIS A006933

Classifier	Definition / Description	Examples / OEIS
Fibonacci-base palindrome	A <i>Fibonacci-base palindrome</i> is a number whose Zeckendorf Fibonacci-base bitstring reads the same forwards and backwards. The Fibonacci-base bitstring is derived from the Zeckendorf decomposition by writing one bit for each Fibonacci number F_k, from the largest index used down to F_2: • 1 indicates that F_k is included in the decomposition, • 0 indicates that it is not. A number is Fibonacci-base palindromic if this bitstring is symmetric.	12: $12 = 8 + 3 + 1 = F_6 + F_4 + F_2$ The corresponding Fibonacci-base bitstring is: bits($F_6..F_2$) = 10101 The bitstring reads the same forwards and backwards $\Rightarrow$ 12 is a Fibonacci-base palindrome. OEIS N/A
Graham's number suffix	Graham's number is an extraordinarily large integer that arose as an upper bound in a problem in Ramsey theory. Although the complete number is far too large to write down, its final decimal digits are known. This classifier recognizes numbers that match the last k decimal digits of Graham's number, where k is the number of digits in the input. NumClass currently supports suffixes up to 10,000 decimal digits.	387: The last three digits of Graham's number are 387. $\Rightarrow$ 387 are Graham's number suffixes. OEIS: N/A
Harshad in all bases	A number is a <i>Harshad number</i> in all bases if it is divisible by the sum of its digits in every base $b \geq 2$ when written in that base. Formally, an integer n is <i>all-base Harshad</i> if: $n \equiv (\text{moddigit-sum}_b(n))$ for every base $b \geq 2$. Only a few numbers satisfy this for all bases: 1, 2, 3 and 4.	3: Check bases $b = 2, 3, 4, 5, \dots$ • Base 2: $3_{10} = 11_2$, digit sum = $1 + 1 = 2$ $3 \equiv 0 \pmod{2} \rightarrow \checkmark$ • Base 3: $3_{10} = 10_3$, digit sum = $1 + 0 = 1$ $3 \equiv 0 \pmod{3} \rightarrow \checkmark$ • Base 4: $3_{10} = 3_4$, digit sum = 3, $3 \equiv 0 \pmod{4} \rightarrow \checkmark$ • Base 5 and higher: $3_{10} = 3_b$, digit sum = 3 $3 \equiv 0 \pmod{b} \rightarrow \checkmark$ $\Rightarrow$ 3 is Harshad in every base. OEIS A005349 (Harshad)
Kaprekar constant (3-digit)	The fixed point of the 3-digit Kaprekar routine (495).	495: Identifies this number as the 3-digit Kaprekar constant, see Kaprekar routine. OEIS A099009
Kaprekar constant (4-digit)	The fixed point of the 4-digit Kaprekar routine (6174).	6174: Identifies this number as the 4-digit Kaprekar constant, see Kaprekar routine. OEIS A099009

Classifier	Definition / Description	Examples / OEIS
LCM-prefix number	An <i>LCM-prefix number</i> is a positive integer that equals the least common multiple of the first k positive integers for some integer $k \geq 1$: $n = \text{lcm}(1, 2, \dots, k)$ Equivalently, it is the smallest positive integer divisible by every integer from 1 through k . LCM-prefix numbers grow in jumps: the value only increases when a new prime or a higher prime power is required. As a result, the same number may equal $\text{lcm}(1..k)$ for several consecutive values of k .	2520 : $2520 = 2^3 \times 3^2 \times 5 \times 7$ No higher prime powers are needed until $k = 11$, so the value remains unchanged for multiple prefixes: $\text{lcm}(1..9) = \text{lcm}(1..10) = 2520$ OEIS A003418
Left factorial	The left factorial, introduced by Đuro Kurepa, is defined as the sum of all factorials less than n !: $!n = 0! + 1! + 2! + \dots + (n-1)!$ A left factorial number is any value occurring in this sequence.	34 : $!5 = 0! + 1! + 2! + 3! + 4!$ $= 1 + 1 + 2 + 6 + 24 = 34$ $\Rightarrow 34$ is a left factorial number. First terms: 1, 1, 2, 4, 10, 34, 154, 874, ... OEIS A003422
Leyland number	A Leyland number is an integer that can be written in the form $x^y + y^x$, where x and y are integers greater than 1. Because the expression is symmetric, $x^y + y^x = y^x + x^y$, NumClass reports only one ordering, normally with $x \leq y$.	32 : $32 = 2^4 + 4^2 = 16 + 16$ $\Rightarrow 32$ is a Leyland number. OEIS: A076980
Lucky number of Euler (Heegner number)	Checks if n is a 'lucky number of Euler' (Heegner number). These are the unique positive integers n for which $Q(\sqrt{-n})$ has class number 1, giving rise to the near-integer phenomenon $e^{\pi\sqrt{n}} \approx \text{integer}$.	43 : $e^{\pi\sqrt{43}} \approx$ 884736743.999777... rounded: 884736744 OEIS A003173
Munchausen number	Equals the sum of its digits, each raised to the power of itself. 0^0 is undefined as a real-number limit, but in digit functions we need every digit to have a well-defined self-power. Two conventions exist: $0^0=1$ (classical): only 1 and 3435 exists. $0^0=0$ (modern digit-function rule), 1, 3435 and 438579088 exists. NumClass uses the modern digit rule.	3435 : Take its digits: 3, 4, 3, 5. Compute the sum of each digit raised to its own power: $3^3 = 27$, $4^4 = 256$, $3^3 = 27$, $5^5 = 3125$ Added: $27 + 256 + 27 + 3125 = 3435$ The sum equals the original number $\Rightarrow 3435$ is a Münchhausen number. OEIS A046253
Octal-interpretable number	Decimal number that represents a valid octal numeral when read as digits in base 8.	773 : Base 10 number 773 can be interpreted an octal numeral: $3 \times 8^0 + 7 \times 8^1 + 7 \times 8^2 = 3 + 56 + 448 = 507$ in base 10. OEIS N/A

Classifier	Definition / Description	Examples / OEIS
Pandigital number (0–9)	Contains each digit 0 through 9 exactly once.	8376201459 : each digit 0-9 appears only once. OEIS A050278
Pandigital number (1–9)	Contains each digit 1 through 9 exactly once.	918573624 : each digit 1-9 appears only once. OEIS A050289
Polydivisible number (magic numbers)	A <i>polydivisible number</i> is a positive integer whose prefixes are successively divisible by their length. Let the decimal digits of a number be $d_1 d_2 \dots d_k$. For each prefix formed by the first i digits: $(d_1 d_2 \dots d_i)$ is divisible by i for all $i = 1, 2, \dots, k$. In other words, when reading the number from left to right, each growing prefix is evenly divisible by the number of digits it contains.	360: Check each prefix: • 3 is divisible by 1 • 36 is divisible by 2 • 360 is divisible by 3 Since all prefixes satisfy the divisibility condition, 360 is polydivisible. OEIS A144688
Riesel candidate	A Riesel number is an odd positive integer n such that $n \cdot 2^k - 1$ is composite for every positive integer k . NumClass first tests exponents up to the configured limit. If a prime value is found, the number is rejected. If no prime is found, the number is a candidate. NumClass then tries to prove the property by finding a covering set of primes that divides $n \cdot 2^k - 1$ for every possible k . If such a covering set is found, the Riesel property is proven. Otherwise the number remains an unresolved candidate.	509203 : The smallest known Riesel number. For this number, a covering set proves that $509203 \cdot 2^k - 1$ is composite for all k . OEIS A101036 (for proven numbers)
Sierpiński candidate	A Sierpiński number is an odd positive integer n such that $n \cdot 2^k + 1$ is composite for every positive integer k . NumClass first tests exponents up to the configured limit. If a prime value is found, the number is rejected. If no prime is found, the number is a candidate. NumClass then tries to prove the property by finding a covering set of primes that divides $n \cdot 2^k + 1$ for every possible k . If such a covering set is found, the Sierpiński property is proven. Otherwise the number remains an unresolved candidate.	78557 : The smallest known Sierpiński number. For this number, a covering set proves that $78557 \cdot 2^k + 1$ is composite for all k . OEIS A076336 (for proven numbers)
Smarandache–Wellin number	Formed by concatenating the first n prime numbers written in base 10.	2357 : The first 4 primes 2, 3, 5, 7 concatenated. OEIS A019518

Classifier	Definition / Description	Examples / OEIS
Strobogrammatic number	Looks the same when rotated 180°, using digits like 0, 1, 6, 8, and 9.	68189: Reads the sum up-side-down. OEIS A000787
Vampire number	Composite number factorable into two equal-length factors containing exactly the same digits as the original number.	1260: $1260 = 21 \times 60$, possible “fangs”: 21 and 60. Check digit lengths: • 1260 has 4 digits • Fangs must have 2 digits each → ✓ • 21 and 60 both have 2 digits → ✓ Check digit multiset Digits of 1260: {1, 2, 6, 0} Digits of 21 and 60 combined: • 21 → {2, 1}, 60 → {6, 0} • Combined → {1, 2, 6, 0} They match exactly → ✓ Zero rule: Fangs may not both end in 0. 21 ends with 1, 60 ends with 0 Only one ends with zero → ✓ ⇒ 1260 is a vampire number with fangs 21 and 60. OEIS A014575

Named Sequences

Numbers that occur in named integer sequences with recursive or combinatorial definitions.

Classifier	Definition / Description	Examples / OEIS																									
2-term Zeckendorf number	A 2-term Zeckendorf number is a number whose Zeckendorf decomposition consists of exactly two Fibonacci numbers. Because Zeckendorf decompositions use only non-consecutive Fibonacci numbers, such a number can be written uniquely as: $n = F_a + F_b$ with $a \geq b + 2$	11: $11 = 8 + 3 = F_6 + F_4$ The Zeckendorf decomposition uses exactly two Fibonacci terms, so 11 is a 2-term Zeckendorf number. OEIS N/A																									
Busy Beaver number (Rado's sigma function)	Maximum number of steps that a halting Turing machine with n states can execute before halting. BB(n) is noncomputable.	6: Maximum number of steps for a 2-state, 2-symbol Busy Beaver that achieves it. (blank = 0, start in state A at cell 0; symbols {0,1}; halting state H): <table><tr><td>State</td><td>Read</td><td>Write</td><td>Move</td><td>Next</td></tr><tr><td>A</td><td>0</td><td>1</td><td>R</td><td>B</td></tr><tr><td>A</td><td>1</td><td>1</td><td>L</td><td>B</td></tr><tr><td>B</td><td>0</td><td>1</td><td>L</td><td>AH</td></tr><tr><td>B</td><td>1</td><td>1</td><td>R</td><td></td></tr></table> Execution trace (from blank tape): - A reads 0 → write 1, R, go B (head to +1) - B reads 0 → write 1, L, go A (head to 0) - A reads 1 → write 1, L, go B (head to -1) - B reads 0 → write 1, L, go A (head to -2) - A reads 0 → write 1, R, go B (head to -1) - B reads 1 → write 1, R, halt (head to 0) It halts after 6 steps, so S(2) = 6, and it leaves 4 ones on the tape, so Σ(2) = 4. OEIS A060843	State	Read	Write	Move	Next	A	0	1	R	B	A	1	1	L	B	B	0	1	L	AH	B	1	1	R	
State	Read	Write	Move	Next																							
A	0	1	R	B																							
A	1	1	L	B																							
B	0	1	L	AH																							
B	1	1	R																								
Carol number	A Carol number is a special kind of number defined by the formula $C_n = 2^n - 2^{\lfloor n/2 \rfloor} - 1$ for integer $n \geq 1$, meaning it's 2 less than the square of one less than a power of two.	47: For $n = 3$: $C_3 = 4^3 - 2^4 = 64 - 16 - 1 = 47$ ⇒ 47 is a Carol number. OEIS A093112																									
Cullen number	Number of the form $n = k \cdot 2^k + 1$ for some integer $k \geq 1$.	25: Take $k = 3$: $C_3 = 3 \times 2^3 + 1 = 3 \times 8 + 1 = 25$ ⇒ 25 is a Cullen number. OEIS A002064																									

Classifier	Definition / Description	Examples / OEIS
Erdős–Woods number	Length of an interval in which every integer shares a non-trivial common factor with the first or last element. k is Erdős–Woods if there are integers a and $b = a + k$ such that for every $a < n < b$ we have $\gcd(n, a) > 1$ or $\gcd(n, b) > 1$.	16: Endpoints: $a = 2184 = 2^3 \times 3 \times 7 \times 13$ $b = 2200 = 2^3 \times 5^2 \times 11$ Property: For every integer n with $2184 < n < 2200$, we have: • either $\gcd(n, 2184) > 1$, • or $\gcd(n, 2200) > 1$. $\Rightarrow 16$ is an Erdős–Woods number. OEIS A059756
Factorial number	Integer that equals $k!$ for some integer $k \geq 0$.	720: $720 = 6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$, $720 \div 2 \div 3 \div 4 \div 5 \div 6 = 1$ $\Rightarrow 720$ is a factorial number. OEIS A000142
Fibonacci number	Member of the sequence defined by $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$.	34: Fibonacci sequence: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, ... $\Rightarrow 34$ is the 9 th Fibonacci number. OEIS A000045
Hamming number (regular number) (Ugly number) (5-smooth number)	Number whose prime factors (and are limited to 2, 3, and 5.	90: $90 = 2 \times 3^2 \times 5$, contains no other prime factors than 2, 3 or 5 $\Rightarrow 90$ is a Hamming number. OEIS A051037
Keith number (Repfigit number)	Number that appears in its own digit-sequence recurrence: each term is the sum of the previous d digits, where d is the number of digits in n . Let n have decimal digits $d_1, d_2, \dots, d_k$. Form a sequence starting with these digits and for all later terms: $a_m = a_{m-1} + a_{m-2} + \dots + a_{m-k}$. If the number n itself appears later in this sequence, then n is a Keith number. Keith numbers are very rare because the digit-based recurrence almost never matches the number again.	197: start with the digits: 1, 9, 7. Initial terms: $a_1 = 1, a_2 = 9, a_3 = 7$ Next: $a_4 = a_1 + a_2 + a_3 = 1 + 9 + 7 = 17$ Next: $a_5 = a_2 + a_3 + a_4 = 9 + 7 + 17 = 33$ Next: $a_6 = a_3 + a_4 + a_5 = 7 + 17 + 33 = 57$ Next: $a_7 = a_4 + a_5 + a_6 = 17 + 33 + 57 = 107$ Next: $a_8 = a_5 + a_6 + a_7 = 33 + 57 + 107 = 197$ Now the sequence looks like: 1, 9, 7, 17, 33, 57, 107, 197, ... And since the sequence reaches 197 $\Rightarrow 197$ is a Keith number. OEIS A007629

Classifier	Definition / Description	Examples / OEIS
Lucas number	A <i>Lucas number</i> is a number in the Lucas sequence, which is very similar to the Fibonacci sequence but starts with different initial values. The Lucas numbers L_n are defined by: $L_0 = 2, L_1 = 1,$ $L_n = L_{n-1} + L_{n-2} \text{ for } n \geq 2.$ So the first values are: 2, 1, 3, 4, 7, 11, 18, 29, 47, 76, ...	18: Take $n = 6$: $L_6 = L_5 + L_4 = 11 + 7 = 18$ $\Rightarrow 18$ is the 6th Lucas number. OEIS A000032
Lucky number	Generated by the “lucky sieve”: repeatedly removing every k^{th} number from the natural numbers. Start with a list of all positive odd integers: 1, 3, 5, 7, 9, 11, 13, 15, 17, ... Because the second number in the list is 3, remove every 3rd <u>position</u> (not every number divisible by 3): 1, 3, 7, 9, 13, 15, 19, ... next surviving number is 3, remove every 3rd remaining number position. 1, 3, 7, 9, 13, 15, 21 next surviving number is 7, remove every 7th number position and continue infinitely. A number that is never removed is called a <i>Lucky number</i>.	13: The sequence left after the sieve is: 1, 3, 7, 9, 13, 15, 21, 25, 31, 33, 37, ..., 13 is in the sequence. $\Rightarrow 13$ is a Lucky number. OEIS A000959
Padovan number	The Padovan sequence $P(n)$ is defined as: $P(0) = P(1) = P(2) = 1$, $P(n) = P(n-2) + P(n-3)$ for $n \geq 3$. Resulting in the following sequence: 1, 1, 1, 2, 2, 3, 4, 5, 7, 9, 12, 16, 21, ...	21: $P(12) = P(10) + P(9) = 12 + 9 = 21$ $\Rightarrow 21$ is a Padovan number. OEIS A000931
Pell number	A <i>Pell number</i> is defined recursively by $P_n = 2P_{n-1} + P_{n-2}$, with $P_0 = 0$, $P_1 = 1$, forming the sequence: 0, 1, 2, 5, 12, 29, 70, 169, 408, 985, ...	12: $P_4 = 2P_3 + P_2 = 2 \times 5 + 2 = 12 \Rightarrow 12$ is the 4th Pell number OEIS A000129
Prime-count landmark	A prime-count landmark is a number equal to $\pi(10^n)$, where $\pi(x)$ is the prime-counting function (the number of primes less than or equal to x). In other words, it is the number of primes not exceeding a power of ten. These values serve as important milestones in computational number theory and are frequently used to verify prime-counting algorithms.	168: $\pi(10^3) = 168$ $\Rightarrow 168$ is a prime-count landmark. OEIS A006880

Classifier	Definition / Description	Examples / OEIS
Sparse Zeckendorf representation	A number has a <i>sparse Zeckendorf representation</i> if its Zeckendorf decomposition uses only a small number of Fibonacci terms. The maximum allowed number of terms is configurable. If the number of Fibonacci terms in the Zeckendorf decomposition does not exceed this limit, the number is classified as <i>sparse</i> .	12: For a maximum of 3 terms: $12 = 8 + 3 + 1 = F_6 + F_4 + F_2$ The decomposition uses three Fibonacci terms $\Rightarrow$ 12 has a sparse Zeckendorf representation. OEIS N/A
Taxicab number (Hardy–Ramanujan number)	Can be expressed as the sum of two positive cubes in at least two distinct ways.	1729: $1729 = 1^3 + 12^3 = 9^3 + 10^3$. OEIS A011541
Tribonacci number	Each term equals the sum of the previous three terms, starting with 0, 0, 1, so the first values are 0, 1, 1, 2, 4, 7, 13, 24, 44, 81, 149, ...	44: $T_8 = T_5 + T_6 + T_7 = 7 + 13 + 24 = 44 \Rightarrow 44 = \text{Tribonacci number } T_8$. OEIS A000073
Ulam number	Term of the Ulam sequence: each is the smallest integer that can be written uniquely as a sum of two distinct earlier terms in exactly one way, starting with 1 and 2. Resulting in the following sequence: 1, 2, 3, 4, 6, 8, 11, 13, 16, 18, 16, ...	6: $6 = 2 + 4$, so 6 can be written as a sum of two earlier Ulam numbers and no other distinct pair of earlier Ulam numbers gives 6. $\Rightarrow 6$ is an Ulam number. OEIS A002858
Woodall number (Riesel number)	Number of the form $n = k \cdot 2^k - 1$ for some integer $k \geq 1$.	63: Take $k=4$: $W_4 = 4 \times 2^4 - 1 = 416 - 1 = 63 \Rightarrow 25$ is a Woodall number. OEIS A003261

Polygonal and Figurate Numbers

Numbers representable as geometric shapes such as triangles, squares, or polygons.

Classifier	Definition / Description	Examples / OEIS
Centered hexagonal number	Number of the form $n = 3 \cdot k(k - 1) + 1$ for some integer $k \geq 1$; represents a centered hexagon pattern.	37: Take $k=4$: $H_4 = 3 \times 4 \times (4-1) + 1 = 37$ $\Rightarrow 37$ is a Centered hexagonal number. OEIS A003215
Centered square number	Number of the form $n = 2 \cdot k(k - 1) + 1$ for some integer $k \geq 1$; geometrically forms a centered square (an odd square).	41: Take $k=5$: $S_4 = 2 \times 5 \times (5-1) + 1 = 41 \Rightarrow 41$ is a Centered square number. OEIS A001844
Centered triangular number	Number of the form $n = (3/2) \cdot k(k - 1) + 1$ for some integer $k \geq 1$.	46: Take $k=6$: Take $k=6$: $T_4 = 3/2 \times 6 \times (6-1) + 1 = 46$ $\Rightarrow 46$ is a Centered triangular number. OEIS A005448

Classifier	Definition / Description	Examples / OEIS
Harshad hexagonal number (intersection)	Hexagonal number that is also divisible by the sum of its decimal digits. $H_n = n(2n - 1)$ $H_n \div \text{digit_sum}(H_n) \in \mathbb{Z}$.	190: Take $n = 10$: $H_{10} = 10(2 \times 10 - 1) = 10 \times 19 = 190$. Digit sum: $1 + 9 + 0 = 10$. $190 \div 10 = 19$ (integer) $\Rightarrow 190$ is both hexagonal and a Harshad number. OEIS N/A
Harshad triangular number (intersection)	Triangular number that is also divisible by the sum of its decimal digits. $T_n = \frac{k(k+1)}{2}$	36: Take $n=8$ $T_8 = \frac{8 \times 9}{2} = 36$. Digit sum: $3 + 6 + 9 = 18$. Check the Harshad property: $36 \div 9 = 4$ (integer) $\Rightarrow 36$ is both triangular and a Harshad number. OEIS A076713
Hexagonal number	Number of the form $n = k(2k - 1)$ for some integer $k \geq 1$.	66: Take $k=6$, $n = 6(2 \times 6 - 1) = 66$ $\Rightarrow 66$ is a hexagonal number OEIS A000384
Lehmer number	Number equal to $(a^k - 1) / (a - 1)$ for integers $2 \leq a \leq 10$, $k > 1$; corresponds to k consecutive 1's in base a .	31: Take $a=2$, $k=5$, $n = (2^5 - 1) / (2 - 1) = 31$ $\Rightarrow 31$ is a Lehmer number. OEIS N/A
Palindromic triangular number (intersection)	Triangular number of the form $T_n = \frac{k(k+1)}{2}$ whose decimal representation is a palindrome.	171: Take $k=18$, $T_n = \frac{18(18+1)}{2} = 171$ and is a palindrome. OEIS A003098
Pentagonal number	Pentagonal number of the form $P_n = \frac{k(3k-1)}{2}$ for some integer $k \geq 1$.	22: Take $k=4$, $P_n = \frac{4(3 \times 4 - 1)}{2} = 22$ $\Rightarrow 22$ is a pentagonal number OEIS A000326
Pronic number (Oblong number) (Heteromecic number)	Number of the form $n = k(k + 1)$; the product of two consecutive integers.	4: $4(4+1) = 4 \times 5 = 20 \Rightarrow 4$ is a Pronic number. OEIS A002378
Repunit	Number consisting solely of digit 1, e.g., 1, 11, 111, etc.	111: Number solely of digit 1 $\Rightarrow 111$ is a repunit. OEIS A002275
Star number (Centered hexagram number) (Centered 12-gonal number)	Number of the form $S_n = 6k(k - 1) + 1$, $k \geq 1$. Represents a centered hexagram or 12-gonal pattern.	3: $S_3 = 6 \times 3 \times 2 + 1 = 36 + 1 = 37$. $\Rightarrow 37$ is a star number. OEIS A003154

Classifier	Definition / Description	Examples / OEIS
Tetrahedral number	Number of the form $T_n = \frac{k(k+1)(k+2)}{6}$ Represents a 3D pyramid with a triangular base.	4: $T_4 = \frac{4 \times 5 \times 6}{6} = \frac{120}{6} = 20.$ $\Rightarrow 20$ is a tetrahedral number. This corresponds to a tetrahedron made of: • 1 point on top • 3 points in the second layer • 6 in the third • 10 in the base And $1 + 3 + 6 + 10 = 20$. OEIS A000292
Triangular number	Number of the form $T_n = \frac{k(k+1)}{2}$ for some integer $k \geq 1$.	15: $5(5+1) \div 2 = 15$ OEIS A000217

Primes and Prime-related Numbers

Numbers that are prime or defined through relationships to prime forms or patterns.

Classifier	Definition / Description	Examples / OEIS
Absolute prime (Permutable prime)	Prime that remains prime under all permutations of its digits.	113: 113, 131, 311 are all prime. OEIS A003459
Automorphic prime (intersection)	Prime whose square ends with the same digits as the prime itself.	5: $5^2 = 25$ and 5 is prime. OEIS A052228
Balanced prime	A prime that is exactly halfway between the previous and the next primes.	53: Primes before and after 53 are 47 and 59, average $(47+59) \div 2 = 53$, the gap between both primes are equal (6) $\Rightarrow 53$ is a balanced prime. OEIS A006562
Bell prime (intersection)	Prime that divides a Bell number, which counts set partitions.	877: Is a Bell number and a prime. OEIS A051130
Both-truncatable prime	Prime that remains prime when digits are removed from either the left or the right.	313: 313, 31, 13 and 3 are all primes. OEIS A020994
Brazilian prime (intersection)	Prime representable as a repdigit in some base b with $2 \leq b \leq n-2$.	8191: Is a base-2 repunit 111111111111_2 and prime. OEIS A377886
Catalan prime	Prime that is also a Catalan number. (See Catalan number for definition) The only prime Catalan numbers are 2 and 5.	5: Is a Catalan number and a prime. OEIS A000108
Chen prime	Prime p for which $p+2$ is either prime or a semiprime (product of two primes).	89: $89+2=91=7 \times 13$, a semiprime. 89 is also a prime. OEIS A109611
Circular prime	Prime that remains prime under all cyclic rotations of its digits.	197: 197, 971 and 791 are all prime. OEIS A068652

Classifier	Definition / Description	Examples / OEIS
Cousin prime	Prime that forms a pair with another prime differing by 4.	967 : 967 and 971 differ by 4 $\Rightarrow$ 967 and 971 are Cousin primes. OEIS A160440 (Smaller member) OEIS A046132 (Larger member)
Cullen prime (intersection)	Prime of the form $p = k \cdot 2^k + 1$ for some integer $k \geq 1$.	393050634124102232869567034555427371542904833 : Let $k = 141$, $141 \times 2^{141} + 1 = p$ and is also prime. OEIS A050920
Cyclops prime (intersection)	Prime with an odd number of digits and a single zero in the middle.	12049 : Odd number with single zero in the middle and is prime. OEIS A134809
Disarium prime (intersection)	Prime that also satisfies the Disarium property (sum of $\text{digits}^{\text{position}} = n$). Only 5 Disarium primes exist: 2, 3, 5, 7 and 89.	89 : $89 = 8^1 + 9^2 = 8 + 81$ and 89 is also prime. OEIS N/A
Emirp	Prime that yields a different prime when its digits are reversed. (The word emirp = prime in reverse).	13 : 13 and 31 are prime. OEIS A006567
Factorial prime	Prime of the form $k! + 1$ or $k! - 1$ for some integer $k \geq 1$.	5039 : $5039 = 7! - 1 \Rightarrow 5039$ is a factorial prime OEIS A002981 ($k! + 1$) OEIS A002982 ($k! - 1$)
Factorion prime (intersection)	Prime that equals the sum of the factorials of its digits. 2 is the only factorion prime.	2 : $2! = 2$ and 2 is a prime. OEIS N/A
Fermat prime	Prime of the form $F_n = 2^{2^n} + 1, n \geq 0$. No Fermat primes with $n \geq 5$ are known.	65537 : Take $n=4$: $F_4 = 2^{2^4} + 1 = 2^{16} + 1 = 65\,536 + 1 = 65\,537$ and is also prime. $\Rightarrow 65537$ is a Fermat prime F_4 . OEIS A019434
Fibonacci prime (intersection)	Prime that is also a term of the Fibonacci sequence: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, ...	89 : 89 is in the Fibonacci sequence and is prime. OEIS A005478
Gaussian prime	Prime element in the ring of Gaussian integers $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$.	13 : Take $3 + 2i$, it's norm is $N(3+2i) = 3^2 + 2^2 = 9 + 4 = 13$ and 13 is a rational prime, the Gaussian integer is irreducible $\Rightarrow 13$ is a Gaussian prime. OEIS A055025

Classifier	Definition / Description	Examples / OEIS
Good prime	A prime p_n is a <i>good prime</i> if it dominates <i>all</i> symmetric products of earlier and later primes: $p_n^2 > p_{n-i} \times p_{n+i}$ for all $1 \leq i \leq n-1$. Equivalently, the local prime gaps around p_n grow “faster than expected” when compared to all previous symmetric gaps.	29: it's index is $n = 10$, and checks show: $29^2 > 2331$, $29^2 > 19 \times 37$, $29^2 > 17 \times 41$, ... holding for all valid i. Thus 29 satisfies the global good-prime condition. OEIS A028388
Good prime (local)	A prime p_n is a <i>local good prime</i> if it exceeds the geometric mean of its immediate neighbors: $p_n^2 > p_{n-1} \times p_{n+1}$ This is the nearest-neighbor version of the good-prime condition. Every global good prime is automatically a local good prime, but the converse need not hold.	17: $17^2 = 289$ and $13 \times 19 = 247$, and since $289 > 247$, the condition is satisfied. OEIS A046869
Happy palindromic prime (intersection)	Prime that is both happy and palindromic in decimal representation.	383: • Happy: repeatedly summing the squares of its digits eventually reaches 1. • Palindromic: Forward and backward reads the same. • Prime: 383 is a prime number. OEIS A364479
Happy prime (intersection)	Prime that remains after repeatedly summing squares of digits until reaching 1.	79: • Happy: repeatedly summing the squares of its digits eventually reaches 1. • Prime: 383 is a prime number. OEIS A035497
Harshad prime (intersection)	Prime in base 10 that is also divisible by the sum of its digits. The only Harshad primes are: 2, 3, 5 and 7.	7: $7 \div 7 = 1$ The only divisors of a prime are 1 and n, so no Harshad primes (base 10) can exist with 2 or more digits. OEIS N/A
Isolated prime	Prime not adjacent to any twin prime (neither $p-2$ nor $p+2$ is prime).	89: 89 is not divisible by 2, 3, 5 and 7 (divisibility up to $\sqrt{89}$) $89 - 2 = 87$, $89 + 2 = 91$, both 87 and 91 are not prime. OEIS A007510

Classifier	Definition / Description	Examples / OEIS
Keith prime	Prime that is also a Keith number under the base-10 digit-recurrence definition. Only 6 Keith primes are known. All base-10 Keith numbers up to 36 digits have been determined and none of those are prime.	197 : It has 3 digits, so $k = 3$. Start the sequence with its digits: $a_1 = 1, a_2 = 9, a_3 = 7$ Now build the sequence, each term = sum of previous 3: • $a_4 = a_1 + a_2 + a_3 = 1 + 9 + 7 = 17$ • $a_5 = a_2 + a_3 + a_4 = 9 + 7 + 17 = 33$ • $a_6 = a_3 + a_4 + a_5 = 7 + 17 + 33 = 57$ • $a_7 = a_4 + a_5 + a_6 = 17 + 33 + 57 = 107$ • $a_8 = a_5 + a_6 + a_7 = 33 + 57 + 107 = 197$ 197 reappears as a_8 and is prime $\Rightarrow$ 197 is a Keith prime. OEIS A048970
Left-truncatable prime	Prime that remains prime after removing leading digits one at a time and no digits are zero.	239 : 239, 39 and 9 are all prime. OEIS A024785
Lucas prime (intersection)	Prime that is also a term in the Lucas sequence: 2, 1, 3, 4, 7, 11, 18, 29, 47, 76, 123, 199, ... (See Lucas number)	199 : 199 is in the Lucas sequence L(11) and is prime. OEIS A005479
Lucky prime (intersection)	Prime that is also a lucky number from the lucky sieve.	223 : $223 = L(43)$ and prime. OEIS A031157
Mersenne prime (intersection)	Prime of the form $2^p - 1$ where p is prime.	8191 : Take $p=13$, $2^{13} - 1 = 8191 \Rightarrow 8191$ is a Mersenne prime. OEIS A000668
Motzkin prime	Prime that is also a Motzkin number. (See Motzkin number). Only 4 Motzkin primes are known.	15511 : 15511 is a Motzkin number and the 3 rd known Motzkin prime. OEIS A092832
Narcissistic prime (intersection)	Prime that is also a narcissistic number (sum of digits ^{count} = n).	28116440335967 : 14 digits, $28116440335967 = 2^{14} + 8^{14} + 1^{14} + 1^{14} + 6^{14} + 4^{14} + 4^{14} + 0^{14} + 3^{14} + 3^{14} + 5^{14} + 9^{14} + 6^{14} + 7^{14}$ and is also prime. OEIS A145380
Padovan prime (intersection)	Prime that is also a Padovan number. (Sequence 1, 1, 1, 2, 2, 3, 4, 5, 7, 9, 12, 16, 21, 28, 37, ...)	37 : 37 is P(14) and is also prime. OEIS A100891
Palindromic prime (intersection)	Prime whose decimal representation reads the same forward and backward.	787 : reads the same forward and backward and is prime. OEIS A002385
Pell prime (intersection)	Prime that is also a Pell number. (See Pell number definition)	29: $P_5 = 2P_4 + P_3 = 2 \times 12 + 5 = 29$ and is also prime. OEIS A096650
Pierpont prime (Class 1 prime)	Prime of the form $2^a \cdot 3^b + 1$ for integers a, b ≥ 0 .	97 : $2^5 \times 3^1 + 1 = 97$ OEIS A005109

Classifier	Definition / Description	Examples / OEIS
Prime triplet member	Member of the only prime triplet P, P+2, P+4 (3, 5, 7).	5: 5 belongs to the prime triplet 3, 5, 7 OEIS N/A
Primorial number	A <i>primorial number</i> is the product of all prime numbers up to a given prime p. The primorial of p is written as $p\#$ and defined as: $p\# = 2 \cdot 3 \cdot 5 \cdots p$. Thus, the sequence begins with: 2, 6, 30, 210, 2310, ... Primorials are the prime analogue of factorials: just as $n! = 1 \cdot 2 \cdot 3 \cdots n$, the primorial multiplies all primes up to p. These numbers grow quickly and appear in many areas of number theory, including prime gaps and sieve methods.	30: $30=2\cdot3\cdot5=5\#$. OEIS A002110
Primorial prime	Prime of the form $p\# \pm 1$, where $p\#$ is the product of the first n primes.	30029: Take $p=13$, $13\# = 2 \times 3 \times 5 \times 7 \times 11 \times 13 = 20030 - 1 = 30029$ and is also prime. OEIS A018239 $p\# + 1$ OEIS A006794 $p\# - 1$ (index)
Proth prime	A <i>Proth prime</i> is a prime of the form $n = k \cdot 2^m + 1$ with k odd and $2^m > k$.	13: Take $k=3$ and $m = 2$ so $2^2 = 4 > 3$, $n = 3 \times 2^2 + 1 = 13$. 13 is prime $\Rightarrow$ 13 is a Proth prime. OEIS A080076
Ramanujan prime	Smallest R satisfying $\pi(x) - \pi(x/2) \geq n$ for all $x \geq R$ where $\pi(x)$ is the prime-counting function. It is known (and tabulated) that the Ramanujan primes begin 2, 11, 17, 29, 41, 47, 59, 67, 71, 97, 101, 107, 127, ...	127: From the sequence $127 = R_{13}$. We must have exactly 13 primes in the interval $(127/2, 127] = (63.5, 127]$. Primes between 64 and 127 are: 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127 There are 13 of them, so $\pi(127) - \pi(63.5) = \pi(127) - \pi(63) = 13$. Thus, at $x = 127$ the inequality holds with equality for $n = 13$. By the known theory and tables of Ramanujan primes, for every $x \geq 127$ we have $\pi(x) - \pi\left(\frac{x}{2}\right) \geq 13,$ and 127 is the smallest integer with this property $\Rightarrow$ 127 is Ramanujan prime R_{13} . OEIS A104272
Repunit prime (intersection)	Prime composed entirely of digit '1's.	11: Entirely composed of the digit 1. OEIS A004022

Classifier	Definition / Description	Examples / OEIS
Right-truncatable prime	Prime that remains prime after removing trailing digits one at a time and no digits are zero.	739 : 739, 73 and 7 are all prime. OEIS A024770
Safe prime	Prime p for which $(p - 1)/2$ is also prime.	47 : 47 is prime, $(47-1)/2 = 33$ and 33 is also prime $\Rightarrow$ 47 is a safe prime. OEIS A005385
Semiprime	Product of exactly two (not necessarily distinct) primes.	33 : $33 = 3 \times 11$ = product of two primes. OEIS A001358
Sexy prime	Prime that forms a pair with another prime differing by 6.	461 : 461 differs by 6 with 467 $\Rightarrow$ 461 is a sexy prime. OEIS A046117 (upper of a pair) OEIS A023201 (lesser of a pair)
Smarandach-Wellin prime (intersection)	Prime composed of the concatenation of first n primes (written in base 10).	23 : First primes 2, 3 and 23 are all prime. OEIS A069151
Sophie Germain prime	Prime p such that $2p + 1$ is also prime.	23 : 23 is prime and $2 \times 23 + 1 = 47$ is also prime. OEIS A005384
Strong prime	Prime p for which $p_{-1} + p_{+1} < 2p$, indicating a narrow prime gap.	11 : Surrounding primes: $7 + 13 = 20 < 2 \times 11 = 22$, $11 > (7 + 13) / 2 \Rightarrow$ Strong prime. OEIS A051634
Super prime (Prime-indexed prime)	Prime whose position in the ordered list of primes is itself prime.	59: 59 is the 17th prime, and 17 is also prime. OEIS A006450
Thin prime (Broughan–Zhou)	A <i>Thin prime</i> is an odd prime p such that $p + 1$ is either a power of 2 or a prime times a power of 2, i.e. $p + 1 = 2^k$ or $p + 1 = q \cdot 2^k$ with q prime, $k \geq 1$.	67 : is prime and odd: $p+1 = 68$. $68 = 17 \times 2^2$ A prime p is thin if: • $p + 1 = 2^k$ (pure power of 2), or • $p + 1 = q \cdot 2^k$ where q is prime. $k = 2, q = 17$ and 17 is prime. So $p + 1$ is a prime times a power of 2. That matches the second case exactly $\Rightarrow$ 67 is a thin prime. OEIS A192869
Triangular prime (intersection)	Prime that is also a triangular number. $T_k = \frac{k(k+1)}{2}$ The only triangular prime number is 3.	3: Take $k = 2$: $T_2 = 2 \times (2 + 1) / 2 = 3 \Rightarrow 3$ is a Triangular prime. OEIS N/A
Twin prime	Prime that forms a pair with another prime differing by 2.	17 : 17 is prime and 19 is also prime. OEIS A001359 (lesser) OEIS A006512 (greater)
Ulam prime (intersection)	Prime that is also a term in the Ulam sequence: 1, 2, 3, 4, 6, 8, 11, 13, 16, 18, 26, 28, 36, 38, 47, ...	47 : 47 is prime and in the Ulam sequence. OEIS A068820

Classifier	Definition / Description	Examples / OEIS
Wieferich prime	Prime p satisfying $2^{(p-1)} \equiv 1 \pmod{p^2}$. Only two are known: 1093 and 3511. If others exist, they must be larger than 2^{64} , Proven by the PrimeGrid project, finished 2022.	1093 : 1093 is prime and $2^{1092} \equiv 1 \pmod{1093^2} \Rightarrow 1093$ is a Wieferich prime (base 2). OEIS A001220
Wilson prime	Prime p satisfying $(p-1)! \equiv -1 \pmod{p^2}$. Only three are known: 5, 13 and 563. Searched up to 2×10^{13} .	563 : 563 is prime and $562! \equiv -1 \pmod{563^2} = 316968 = 563^2 - 1 \Rightarrow 563$ is a Wilson prime. OEIS A007540
Woodall prime (intersection)	Prime of the form $p = k \cdot 2^k - 1$ for some integer $k \geq 1$.	23 : 23 is prime and take $k = 3$: $3 \cdot 2^3 - 1 = 3 \times 8 - 1 = 23$. $\Rightarrow 23$ is a Woodall prime. OEIS A050918
Weak prime	Prime p for which $p_{-1} + p_{+1} > 2p$, indicating a relatively wide prime gap.	13 : Surrounding primes: $11 + 17 = 28 > 2 \times 13 = 26$, $13 < (11+17) / 2 \Rightarrow 13$ is a Weak prime. OEIS A051635

Pseudoprimes and Cryptographic Numbers

Composite numbers satisfying certain modular properties or used in cryptographic theory.

Classifier	Definition / Description	Examples / OEIS
Blum integer	Semiprime $n = p \cdot q$ where p and q are distinct primes satisfying $p \equiv q \equiv 3 \pmod{4}$.	21 : $21 = 3 \times 7$, 3 and 7 are distinct primes. Prime modulo 4: $3 \equiv 3 \pmod{4}$ $7 \equiv 3 \pmod{4}$ because $7 = 4 + 3$ Both prime factors satisfy $p \equiv q \equiv 3 \pmod{4} \Rightarrow 21$ is a Blum integer. OEIS A016105

Classifier	Definition / Description	Examples / OEIS
Carmichael number	A composite number n is <i>Carmichael</i> if it satisfies Fermat's little theorem for every base that's coprime to it: For all a with $\gcd(a, n) = 1$: $a^{n-1} \equiv 1 \pmod{n}$. A convenient way to <i>prove</i> a number is Carmichael is Korselt's criterion: A composite integer n is a Carmichael number iff 1. n is squarefree, and 2. For every prime p dividing n, we have $p - 1 \mid n - 1$.	561: Prime factors: $561 = 3 \times 11 \times 17$ (composite) "squarefree" means no prime square divides it. • $3^2 = 9 \nmid 561$ (since $561/9 = 62.\bar{3}$). • $11^2 = 121 \nmid 561$ (since $561/121 \approx 4.63$). • $17^2 = 289 \nmid 561$ (since $561/289 \approx 1.94$). So no p^2 with $p \in \{3, 11, 17\}$ divides 561. Therefore, 561 is squarefree. Each prime divisor p of 561 must be $p - 1 \mid 561 - 1 = 560$. • For $p = 3$: $p - 1 = 2$. $560/2 = 280$, so $2 \mid 560$. • For $p = 11$: $p - 1 = 10$. $560/10 = 56$, so $10 \mid 560$. • For $p = 17$: $p - 1 = 16$. $560/16 = 35$, so $16 \mid 560$. for every prime divisor p of 561, we have $p - 1 \mid 560$. By Korselt's criterion, this implies: 561 is a Carmichael number. Equivalently: for every integer a with $\gcd(a, 561) = 1$, $a^{560} \equiv 1 \pmod{561}$, even though 561 is not prime. OEIS A002997
Cunningham number	A number of the form $b^n \pm 1$ for integer bases b. These numbers play an important role in factorization and computational number theory and are catalogued in the Cunningham project.	$511 = 2^9 - 1$ is a Cunningham number. OEIS: A080262

Classifier	Definition / Description	Examples / OEIS
Euler–Jacobi pseudoprime	An <i>Euler–Jacobi pseudoprime</i> is a composite odd integer n such that for some base a with $\gcd(a, n) = 1$, $a^{\frac{n-1}{2}} \equiv \left(\frac{a}{n}\right) \pmod{n},$ where $\left(\frac{a}{n}\right)$ is the Jacobi symbol. If n were prime, this would always hold (Euler’s criterion), so when a composite n satisfies it, we call it an Euler–Jacobi pseudoprime to base a in $\{2, 3, 5, 7, 11, 13\}$.	1105: Prime factors: $5 \times 13 \times 17$ $\gcd(2, 1105) = 1$, so 1105 is odd, composite and coprime to 2. The Jacobi symbol is multiplicative: $\left(\frac{2}{1105}\right) = \left(\frac{2}{5}\right)\left(\frac{2}{13}\right)\left(\frac{2}{17}\right).$ Use the formula for odd prime p: $\left(\frac{2}{p}\right) =$ $\begin{cases} +1 & \text{if } p \equiv \pm 1 \pmod{8}, \\ -1 & \text{if } p \equiv \pm 3 \pmod{8}. \end{cases}$ For $p = 5$: $5 \equiv 5 \equiv -3 \pmod{8} \Rightarrow \left(\frac{2}{5}\right) = -1$. - For $p = 13$: $13 \equiv 5 \equiv -3 \pmod{8} \Rightarrow \left(\frac{2}{13}\right) = -1$. - For $p = 17$: $17 \equiv 1 \pmod{8} \Rightarrow \left(\frac{2}{17}\right) = +1$. $\Rightarrow \left(\frac{2}{1105}\right) = (-1) \cdot (-1) \cdot (+1) = +1$. Thus the Jacobi symbol is $\left(\frac{2}{1105}\right) = 1$. $2^{\frac{1105-1}{2}} = 2^{552} \pmod{1105} \equiv 1$ $2^{\frac{1105-1}{2}} \equiv \left(\frac{2}{1105}\right) \pmod{1105}$ while being composite $\Rightarrow$ 1105 is an Euler–Jacobi pseudoprime to base 2. OEIS A047713
Fermat pseudoprime	Composite number n satisfying $a^{n-1} \equiv 1 \pmod{n}$ for at least one base a in $\{2, 3, 5, 7, 11, 13\}$.	341: $341 = 11 \times 31 = \text{composite}$ $2^{340} \pmod{341} = 1$ So although 341 is composite, it passes Fermat’s primality test for base 2 $\Rightarrow$ 341 is a Fermat pseudoprime to base 2. OEIS A001567 (base 2)
Lucas–Carmichael number	Odd, squarefree composite n such that $p + 1$ divides $n + 1$ for every prime divisor p of n.	399: Prime factors: $399 = 3 \times 7 \times 19$, 399 is odd and composite. $n + 1 = 400$ Check divisibility condition: For $p=3$, $p+1=4$, $400/4 = 100$ For $p=7$, $p+1=8$, $400/8 = 50$ For $p=19$, $p+1=20$, $400/20 = 20$ Thus, for every prime divisor p of 399, $p+1$ divides $399+1$ OEIS A006972

Classifier	Definition / Description	Examples / OEIS
RSA challenge number	One of the published RSA Challenge semiprimes (RSA-100, RSA-110, RSA-120)	152260502792253336053561 837813263742971806811496 138068865790849458012296 325895289765400035069200 6139 : = 379752279369436739 228088727554456278545655 36638199 × 40094690950920 881030683735292761468389 2148997240 OEIS N/A
Strong pseudoprime	Composite number n that passes the Miller–Rabin strong probable prime test for at least one base in {2, 3, 5, 7, 11, 13}. For an odd composite n, write $n - 1 = 2^s d$ with d odd. Then n is a strong pseudoprime to base a if • either $a^d \equiv 1 \pmod{n}$, • or for some $0 \leq r < s$, $a^{2^r d} \equiv -1 \pmod{n}$. (If a number passes this test but we don't yet know it's composite, we call it a <i>strong probable prime</i>; if we know it's composite, it's a <i>strong pseudoprime</i>.)	3215031751: $n = 151 \times 751 \times 28351$ (composite) with $d = \frac{n-1}{2} = 1607515875$ $2^d \equiv 1 \pmod{n}$ $3^d \equiv -1 \pmod{n}$ $5^d \equiv 1 \pmod{n}$ $7^d \equiv -1 \pmod{n}$ n passes the strong test for bases 2, 3, 5, and 7 even though it is composite $\Rightarrow$ 3215031751 is a strong pseudoprime in this case to bases 2, 3, 5, 7. OEIS A001262 (base 2) OEIS A020229 (base 3) OEIS A020231 (base 5) OEIS A020233 (base 7)

Appendix B – Profile Settings

Notes on Defaults and Fallback Values

NumClass settings are organized into named sections (e.g. [FACTORING], [DISPLAY_SETTINGS], [CLASSIFIER.ZUMKELLER]).

A key in one section is entirely independent from a key with the same name in another section.

Global default vs. classifier-specific fallback

Most settings have a single default value shown in this appendix.

However, a small number of settings appear in multiple parts of the code, with different fallback defaults depending on the context.

These are rare but intentional:

- The global default shown in the table is the canonical value.
- Some classifiers implement a local fallback (usually smaller or stricter) that is used only when the key is not defined in the profile.

Rule:

If a value is set in the profile, that value overrides *all* fallback defaults everywhere in NumClass.

[_PROFILE_]

This section is used to describe the intention of the profile. (Not used in any settings.)

```
name = "all"
description = "Enable every category/classifier; useful for demos and maximal coverage."
```

[CATEGORIES]

This section defines which classifier category is enabled (true) or disabled (false).

```
"Arithmetic and Divisor-based"      = true
"Combinatorial and Geometric"      = true
"Conjectures and Equation-based"    = true
"Digit-based"                      = true
"Diophantine Representations"      = true
"Dynamical Sequences"              = true
"Fun Numbers"                      = true
"Mathematical Curiosities"         = true
"Named Sequences"                  = true
"Polygonal and Figurate Numbers"    = true
"Primes and Prime-related Numbers" = true
"Pseudoprimes and Cryptographic Numbers" = true
```

[INCLUDE_EXCLUDE]

Specifies specific classifiers to be included or excluded.

Use the classifier label name in uppercase, replace (subsequent) non-alphanumeric characters with an underscore '_'. Example: label "collatz (3N+1) sequence" becomes "COLLATZ_3N_1_SEQUENCE"

If none are specified, all classifiers will be whitelisted.

Specifying the classifiers true will whitelist these classifiers, all others will be False.

Specifying the classifiers false will blacklist these classifiers, all others will be True.

[BEHAVIOUR]

Name	Function	Type	Default
FAST_MODE	FAST_MODE = true - NumClass operates in fast mode, heavy or slow classifiers may be skipped. - Step-timeouts are enforced in long-running routines (e.g., aliquot sequences). - Classifier-specific complexity caps are respected (e.g., maximum n thresholds, recursion depth limits, solver time budgets). - Results remain correct, but not all classifications are attempted. FAST_MODE = false - NumClass operates in full computation mode, slow or expensive classifiers are allowed to run without limits on large integers or time budget. - This can significantly increase runtime, especially for large integers or combinatorial classifiers.	Bool	true
MAX_DIGITS	Maximum allowed number of digits for any integer processed by NumClass.	Int	100000

[FACTORING]

Name	Function	Type	Default
BITLEN_CAP	Maximum bit-length NumClass will attempt to factor. If a number exceeds this size, factoring is skipped to avoid impossible workloads.	Int	10000
MAX_TIME_S	Maximum total time (in seconds) NumClass will spend factoring a single number. If the budget runs out, the remaining part is returned as an unfactored composite.	Float	2.0
SMALL_DIGIT_THRESHOLD	Digit count below which NumClass uses a fast, direct factoring method. Larger numbers use the slower, time-limited “big-n” factoring path.	Int	80
STEP_LIMITS	Workload limits for the successive stages of the big-n factoring process. Higher values allow deeper searches but use more time.	List	[1000, 10000, 100000, 1000000]
USE_ECM	Enable elliptic-curve factoring (ECM) where available. May speed up factoring of certain hard composites at the cost of slightly higher overhead.	Int	False

NOFORK_DIGIT_CAP	Maximum number of decimal digits for direct in-process factorization on platforms that do not support safe multiprocessing (such as iOS/a-Shell). Numbers exceeding this limit are returned as an unfactored composite remainder instead of attempting potentially expensive factorization.	Int	80
NOFORK_LIMIT_CAP	Maximum SymPy factorint() limit used by the no-fork factoring backend. This limits trial division and related factorization methods on platforms where bounded subprocess factoring is unavailable. Higher values may improve factorization results at the expense of longer execution times.	Int	10000

DISPLAY_SETTINGS]

Name	Function	Type	Default
DIVISOR_PRINT_LIMIT	Maximum number of individual divisors of $ n $ that NumClass will list explicitly. If there are more divisors than this limit, the list is truncated with a “showing first ... of ...” note.	Int	100
REPEATING_DEC_PRINT_LIMIT	Maximum number of digits used when previewing the decimal expansion of $1/n$ in base 10. Longer repeating decimals are cut off after these many digits, with an ellipsis if needed; period length logic is unaffected.	Int	60
SEQUENCE_PRINT_LIMIT	Maximum number of terms from each aliquot-style sequence (aliquot, quasi-aliquot, unitary) that are shown in the statistics block. Longer sequences are truncated after these many steps.	Int	50 (display), 300 (dynamic helpers)
SKIPPED_MAX_LINES	Threshold for showing individual skipped classifications. If the number of skipped items is $\leq$ this value, all are listed; if it is larger, only a count is shown. 0 means unlimited.	Int	0
SHOW_ALIQUOT_PROGRESS	Show live, step-by-step progress messages while computing aliquot-type sequences. Turn this off to hide the live “evaluating...” / step trace and only show the final summarized sequence.	Bool	true
SHOW_ALIQUOT_SEQUENCE	Show the standard aliquot sequence of n in the “Divisor statistics of $ n $ ” section. Turn this off if you only want divisor sums and counts without the full $s(n) \rightarrow s(s(n)) \rightarrow \dots$ trajectory.	Bool	true

Name	Function	Type	Default
SHOW_ALIQUOT_TIME	Include the total computation time of the aliquot sequence in its summary line (after the sequence). If disabled, the sequence and peak info are shown, but the elapsed time is omitted.	Bool	true
SHOW_CLASSIFIER_DETAILS	Show a second “Details:” line (or lines) under each classification, with explanations, witnesses, or cycle information. Turn this off for a compact one-line-per-classifier view.	Bool	true
SHOW_COLLATZ_PEAK	Controls how the Collatz ($3n+1$) trajectory peak is displayed in dynamical-sequence output: • "Full" – show the full peak information, • "Shortened" – show an abbreviated summary, • "None" – hide the Collatz peak details entirely.	String	"Full"
SHOW_DIGITAL_ROOT_SEQ	For the “Digital root of $ n $ ” line, show the full digital-root sequence and additive persistence, not just the final root. If disabled, only the final digital root and persistence are printed.	Bool	true
SHOW_DIVISOR_STATS	Enable or disable the entire “Divisor statistics of $ n $ ” section ($\tau(n)$, divisors, $\sigma/s/q$, unitary sums, aliquot sequences, etc.). Turn this off if you only care about basic number statistics and classifications.	Bool	true
SHOW_DIVISORS	Show the explicit list of divisors of $ n $ (up to DIVISOR_PRINT_LIMIT) in the divisor statistics section. If disabled, divisor counts and sums are still shown, but the divisor list itself is hidden.	Bool	true
SHOW_MULTIPLICATIVE_SEQ	For “Mult. persistence”, show the full multiplicative persistence sequence ($n \rightarrow$ product of digits $\rightarrow \dots$) instead of just the final persistence value. Disable to keep only the final count.	Bool	true
SHOW_QUASI_ALIQUOT_SEQUENCE	Show the quasi-aliquot sequence of n , generated by iterating $q(n) = s(n) - 1$. Disable to hide this sequence while keeping the main aliquot sequence.	Bool	true
SHOW_REPEATING_DEC_COUNT	On the $1/n$ (base 10) line, show the bracketed period information like [period k] or [rep $\geq L$] where available. If disabled, you still see the decimal preview, but not the explicit period length / lower bound.	Bool	true

Name	Function	Type	Default
SHOW_SKIPPED	Show the “Skipped classifications due to constraints or settings” footer listing classifiers that were not evaluated (e.g. time-capped, disabled, or size-limited). Turn this off to omit the skipped-classifier list.	Bool	true
SHOW_UNITARY_ALIQUOT_SEQUENCE	Show the unitary aliquot sequence based on $s^*(n)$ (sum of proper unitary divisors) in the divisor statistics section. If enabled together with the normal aliquot sequence, NumClass also notes when the unitary and classical sequences coincide.	Bool	false
SHOW_ZECKENDORF	Enable or disable display of the Zeckendorf decomposition in the <i>Number statistics</i> section.	Bool	true
SHOW_ZECKENDORF_BITS	Show the Fibonacci-base bitstring corresponding to the Zeckendorf decomposition.	Bool	true
ZECKENDORF_MAX_TERMS	Maximum number of Fibonacci terms shown in the Zeckendorf decomposition. If the decomposition uses more terms, a compact summary is displayed instead. A value of 0 disables truncation (show all terms).	Int	20
ZECKENDORF_TRUNC_PREVIEW	When the Zeckendorf decomposition is truncated, show a short preview of the first and last Fibonacci indices used.	Bool	true

[FORMATTING]

Name	Function	Type	Default
ELLIPSIS	Symbol used when abbreviating long numbers (e.g., 1234567 → 123...567).	String	"..."
ELLIPSIS_SPLIT	Symbol used when splitting long sequences during truncation — the “big ellipsis” that appears between head and tail parts of a sequence preview.	String	"..."
NUM_ABBR_ENABLED	Enable abbreviation of large integers in sequences and peak displays. If disabled, all numbers are printed in full, regardless of size.	Bool	true
NUM_ABBR_HEAD	How many leading digits to keep when abbreviating a large integer (before the ellipsis).	Int	10
NUM_ABBR_TAIL	How many trailing digits to keep when abbreviating a large integer (after the ellipsis).	Int	10
NUM_ABBR_THRESHOLD	Minimum digit length at which integer abbreviation activates. Numbers shorter than this threshold are printed in full.	Int	10

REPEATING_DEC_STYLE	Visual style used to mark the repeating part of the decimal expansion of $1/n$: • "paren" $\rightarrow (123)$ • "overline" $\rightarrow \overline{123}$ • "underline" $\rightarrow \underline{123}$ This affects only display; period detection is unchanged.	String	"underline"
REPEATING_DEC_MIN_REP_SHOWN	Minimum number of digits required before NumClass shows a repeating block. Shorter repeats are displayed as non-repeating to avoid noisy output.	Int	12
SEQUENCE_ARROW	Symbol used to join terms in sequences (e.g., $3 \rightarrow 10 \rightarrow 5 \rightarrow 16$). Changing it allows stylistic customization such as " $\rightarrow$ ", " $\Rightarrow$ ", etc.	String	" $\rightarrow$ "
UNITFRAC_MAX_REDUCED	Maximum value allowed when showing certain unit-fraction decompositions in reduced form. Used as a safety cap to avoid producing extremely large denominators in symbolic expansions.	Int	1000000

[OUTPUT]

Output file examples:

OUTPUT_FILE = ""	# No output file
OUTPUT_FILE = "."	# Write results to <n>.txt in the current folder
OUTPUT_FILE = "results/"	# Write results to folder/file: ./results/<n>.txt
OUTPUT_FILE = "log.txt"	# Append all results to a single file in the current folder

Settings used by multiple classifiers and classifier helpers

[ALIQUOT]

Name	Function	Type	Default
MAX_STEPS	Maximum number of steps (transitions) NumClass will follow in the aliquot, quasi-aliquot, or unitary aliquot sequence. The sequence always includes the starting value, so a full run has max_steps + 1 terms.	Int	250 (in aspiring, socially aspiring and sociable classifiers, 50 in divisor stats.
STATS_MAX_PEAK_DIGITS	Maximum allowed digit length of any term in the sequence. If a term grows beyond this size, NumClass stops safely with: "peak digit cap exceeded". Useful for preventing enormous intermediates (e.g. factorial-like explosions) during explorations.	Int	10000000
STATS_MAX_START_DIGITS	Digit-length threshold above which aliquot statistics are skipped. 0 means unlimited.	Int	0
STATS_MAX_TIME_S	A hard limit (in seconds) for the entire sequence. If the total elapsed time exceeds this cap, NumClass stops early with:	Float	0.5

	“sequence time exceeded guard”. This prevents runaway cases (e.g., hard $\sigma(n)$ evaluations) while still allowing deep sequences.		
STEP_TIME_LIMIT	Per-step soft timeout (in seconds) used only when fast mode is enabled. If a single $\sigma(n)$ or $\sigma^*(n)$ computation takes longer than this, the sequence stops early with: “step time exceeded guard”. This lets you explore many small aliquot steps very quickly without accidental slow terms.	Float	0.30
VALUE_LIMIT	Maximum allowed size of terms in aliquot-type sequences used by aspiring/sociable classifiers. If a term exceeds this value, those classifiers stop early and treat the run as guarded.	Int	10000000

[COLLATZ_LIKE]

Name	Function	Type	Default
A_VALUE	Multiplier a used in the odd step of the generalized Collatz rule: $\text{odd} \rightarrow a \cdot n + b$.	Int	
B_VALUE	Additive constant b used in the odd step of the rule: $\text{odd} \rightarrow a \cdot n + b$.	Int	
ALLOW_EVEN_A_OR_B	Allow even values for a or b (default requires both to be odd).	Bool	false
A_B_LIMIT	Maximum allowed absolute value of a and b .	Int	9
DRIFT_BITS_EPS	Average bit-growth threshold used by the divergence guard. If recent odd-macro-steps grow by more than this many bits on average, the sequence is considered diverging and is stopped.	Float	0.25
DRIFT_MIN_BITS	Minimum bit-length before the divergence guard is allowed to trigger. Prevents aborting while values are still small.	Int	4096
DRIFT_WINDOW	Number of recent macro-steps included in the sliding-window average for the divergence detector.	Int	256
MAX_STEPS	Maximum number of iteration steps allowed. If reached, the sequence stops with reason “steps $\geq$ MAX_STEPS”.	Int	200000
TIME_BUDGET_S	Maximum time (in seconds) that any single classifier is allowed to spend on its computation. NumClass stops the computation early when exceeding this budget and reports the classifier as skipped.	float inf	None
VALUE_CAP_BITS	Safety cap on how large values in a generalized Collatz sequence may grow. While A_VALUE and B_VALUE determine the growth rule ($\text{odd} \rightarrow$	Int	20000

	a·n+b), this setting prevents runaway bit-length explosions by aborting the sequence once a term exceeds the allowed size.		
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[COMBINATORIAL]

Name	Function	Type	Default
TIME_BUDGET_S	Maximum wall-clock time (in seconds) that each combinatorial classifier is allowed to spend when fast mode is enabled. If a check takes longer than this, it stops early and may fall back to an OEIS b-file (Bell, Catalan, Motzkin) or just give up.	Int	Float

[DIOPHANTINE]

Name	Function	Type	Default
ALLOW_ZERO_IN_DECOMP	Allow decompositions where one of the components is zero (e.g. $n = a^2 + 0^2$). Affects all sum-of-k squares/cubes classifiers.	Bool	false for sums of 2 cubes/squares and 3 squares; true for sums of 3 cubes.
MAX_ABS_FOR_SUM_OF_3_CUBES	Maximum absolute value allowed for each variable a, b, c when searching solutions to: $n = a^3 + b^3 + c^3$ Larger values drastically increase search space; this caps the cube search box to [-B, B].	Int	100
MIN_PALINDROME_DIGITS	Minimum number of digits a palindrome must have when used in decompositions like: • $n = a + b$ (sum of 2 palindromes) • $n = a + b + c$ (sum of 3 palindromes) If the setting is omitted, no minimum is enforced.	Int	None
MAX_SOL_SUM_OF_2_CUBES	Maximum number of decompositions shown for $n = a^3 + b^3$. Actual search may find more; only the first N canonical solutions are displayed.	Int	20
MAX_SOL_SUM_OF_2_PALINDROMES	Maximum number of decompositions shown for $n = p_1 + p_2$, where p_i are palindromes. If the setting is omitted, show all found (may be many).	Int	None
MAX_SOL_SUM_OF_2_SQUARES	Maximum number of decompositions shown for $n = a^2 + b^2$ (canonical pairs $a \leq b$, $a \geq 0$). Does not affect existence checking — only how many are printed.	Int	20
MAX_SOL_SUM_OF_3_CUBES	Maximum number of decompositions shown for $n = a^3 + b^3 + c^3$. NumClass always injects the known “celebrity solutions” even if capped.	Int	20

Name	Function	Type	Default
MAX_SOL_SUM_OF_3_PALINDROMES	Maximum number of decompositions shown for $n = p_1 + p_2 + p_3$, where p_i are palindromes.	Int	None
MAX_SOL_SUM_OF_3_SQUARES	Maximum number of decompositions shown for $n = a^2 + b^2 + c^2$ (canonical triples). Legendre's conditions still apply before enumeration.	Int	20

Classifier specific settings

[CLASSIFIER.BELL]

Name	Function	Type	Default
MAX_K	Upper bound on the index k when searching for a Bell number $B(k)$. NumClass computes $B(0), B(1), \dots, B(k)$ up to this limit (and the time budget) before giving up and falling back to the OEIS b-file.	Int	2500

[CLASSIFIER.CATALAN]

Name	Function	Type	Default
MAX_K	Maximum index k to consider when checking whether n is a Catalan number C_k . NumClass iterates $C_0, C_1, \dots, C_k$ up to this cap (and the time budget) before giving up and using the OEIS b-file. This is an index, not a value.	Int	200000

[CLASSIFIER.DUCCI]

Name	Function	Type	Default
MAX_STEPS	Maximum number of Ducci iterations NumClass will perform. If this limit is reached, the computation stops and reports that the step cap was hit.	Int	256

[CLASSIFIER.EGYPTIAN]

Used by Egyptian m/n , Erdős–Straus ($4/n$, $k=3$) and Sierpiński ($5/n$, $k=4$)

Name	Function	Type	Default
DISTINCT	Require all denominators to be distinct ($1/d_1 + \dots + 1/d_k$ with all d_i different).	Bool	true
K_VALUE	Number of unit fractions in the decomposition.	Int	2
M_VALUE	Numerator m in the target fraction m/n to be decomposed.	Int	4
MAX_DENOM	Upper bound on denominators d_i in unit fractions ($1/d_i$).	Int	10000000
MAX_EXAMPLES	Maximum number of distinct decompositions to return; only the first is shown in details.	Int	1

[CLASSIFIER.GOLDBACH]

Name	Function	Type	Default
MAX_LIST	Maximum number of Goldbach pairs to display in the details (when listing examples).	Int	10
PROBE_CAP	In fast mode (no sieve, just probing primes), maximum number of candidate primes p to try before giving up.	Int	500000
SHOW_ALL	Whether to list all Goldbach pairs or only up to MAX_LIST.	Bool	false

SHOW_COUNT	Whether to report the total number of Goldbach pairs for n (e.g., “(12 pairs; examples: ...)”).	Bool	True
SIEVE_MAXN	Maximum n for which NumClass will build an explicit sieve to enumerate <i>all</i> Goldbach pairs. For $n > \text{SIEVE_MAXN}$, only the fast-probe path is used.	Int	10000000

[CLASSIFIER.FUBINI]

Name	Function	Type	Default
MAX_INDEX	Maximum index n to consider when checking whether v is a Fubini (ordered Bell) number $F(n)$. NumClass computes $F(0), F(1), \dots, F(n)$ up to this limit (and the time budget) before stopping.	Int	5000

[CLASSIFIER.GRAFTING]

Name	Function	Type	Default
ALLOW_STRADDLE	Allow a “straddle” match, where part of n ’s digits are at the end of the integer part and the rest at the start of the fractional part of $n^{1/p}$.	Bool	true
EARLY_STOP_ZEROS	Early-stop heuristic based on leading zeros in the fractional part of $n^{1/p}$. If $\text{EARLY_STOP_ZEROS} > 0$ and the fractional part starts with at least that many zeros for a sufficiently large n , higher orders are not tried. Set to 0 to disable this heuristic.	Int	2
GUARD_EXTRA_DIGITS	Extra safety margin on the number of fractional digits computed when checking for grafting (beyond the length of n). Higher values may improve robustness but cost more Decimal work.	Int	8
HARD_MAX_ORDER	Absolute upper bound on the exponent p . Even if adaptive logic suggests a larger value, p never exceeds this	Int	12
MAX_DIGITS	Maximum number of digits of n for which the grafting test is attempted. Larger numbers are rejected immediately with a “maximum number of digits exceeded” message.	Int	200
MAX_ORDER	Soft upper bound on the exponent p (root order) to try. The classifier may use a smaller adaptive cap for very small or large numbers of digits.	Int	6

Name	Function	Type	Default
SIDE	Where the digits of n are allowed to appear around the decimal point of $n^{1/p}$: • "left" – only as a suffix of the integer part, • "right" – only as a prefix of the fractional part, • "either" – allow left, right, or a straddle across the decimal point. Invalid values fall back to "either".	String	"either"

[CLASSIFIER.K_FULL]

Name	Function	Type	Default
K_VALUE	Default exponent threshold k for the generic “k-full number” classifier (every prime in the factorization must appear with exponent $\geq k$).	Int	4

[CLASSIFIER.LAH]

Name	Function	Type	Default
MAX_INDEX	Maximum index n to scan when looking for Lah numbers $L(n, k)$ that equal v . For each {MAX_INDEX}, NumClass scans all k in that row; higher values allow more hits but cost more time.	Int	10000

[CLASSIFIER.LCM_PREFIX]

Name	Function	Type	Default
MAX_DIGITS	Safety cap for very large inputs. If the input integer has more than this many decimal digits, the LCM-prefix classifier skips it (to avoid expensive factorization on huge numbers).	Int	2000
MAX_K	Safety cap for the inferred witness value k . If the candidate k is larger than this limit, the classifier skips the “check all primes $q \leq k$ ” step (which can be slow for very large k).	Int	500000

[CLASSIFIER.LEGENDRE]

Name	Function	Type	Default
MAX_LIST	Maximum number of primes between n^2 and $(n+1)^2$ to display.	Int	20
SHOW_ALL	Whether to list all primes in the interval $(n^2, (n+1)^2)$ or only up to MAX_LIST.	Bool	false
SHOW_COUNT	Whether to show the total number of primes in that interval.	Bool	true

[CLASSIFIER.LEMOINE]

Name	Function	Type	Default
MAX_LIST	Maximum number of (p, q) pairs to display.	Int	10

SHOW_ALL	Whether to list all representations $n = p + 2q$ or only up to MAX_LIST.	Bool	false
SHOW_COUNT	Whether to include the total number of Lemoine representations.	Bool	true

[CLASSIFIER.MOTZKIN]

Name	Function	Type	Default
MAX_INDEX	Maximum index n to consider when checking whether v is a Fubini (ordered Bell) number $F(n)$. NumClass computes $F(0), F(1), \dots, F(n)$ up to this limit (and the time budget) before stopping.	Int	5000
MAX_K	Maximum index k to consider when checking whether n is a Motzkin number M_k . NumClass walks the Motzkin recurrence up to this index (and the time budget) and then falls back to the OEIS b-file if needed.	Int	200000

[CLASSIFIER.POWER_FAMILY]

Used by Sierpinski and Riesel candidate and Leyland number

Name	Function	Type	Default
MAX_EXP	Maximum exponent k examined when searching for a prime witness of the form: $n \cdot 2^k \pm 1$. If a prime is found for some $1 \leq k \leq \text{MAX_EXP}$, the number is not a Sierpiński or Riesel candidate. Higher values increase the search range but also increase computation time.	Int	2000
COVER_PRIME_LIMIT	Maximum prime value used during covering-set searches. Only covering primes satisfying $p \leq \text{COVER_PRIME_LIMIT}$ are considered. Increasing this value may discover additional covering sets at the expense of longer runtimes.	Int	500
COVER_MODULUS_CAP	Maximum modulus examined during covering-set searches. Only congruence classes with modulus values up to COVER_MODULUS_CAP are tested. Larger values increase the search space and may reveal additional covering relations.	Int	5000
COVER_TIME_BUDGET_S	Maximum wall-clock time allowed for a covering-set search. When the time budget is exhausted, the search stops and the property remains unresolved if no covering set has been found.	Float	1.0

[CLASSIFIER.PRIMORIAL]

Name	Function	Type	Default
MAX_DIGITS	Maximum decimal digits of n for which the primorial check is attempted; larger inputs are skipped.	Int	200
MAX_PRIMES	Maximum number of successive primes the primorial test may multiply; exceeded when p grows too large.	Int	5000

[CLASSIFIER.RANDA] (Reverse-AND-Add)

Name	Function	Type	Default
MAX_STEPS	Maximum number of reverse-and-add iterations NumClass will perform when building the sequence ($n \rightarrow n + \text{reverse}(n)$). If no palindrome is found within MAX_STEPS, the classifier stops and reports that no palindrome was reached within the cap.	Int	1000
PRINT_ALWAYS_IF	If the number of steps needed to reach a palindrome is $\leq$ this value, NumClass prints the full sequence, even if global sequence limits would normally truncate it. Sequence truncation still respects SEQUENCE_PRINT_LIMIT	Int	30

[CLASSIFIER.ULAM]

Name	Function	Type	Default
TIME_BUDGET_S	Maximum time (in seconds) that any single classifier is allowed to spend on its computation. NumClass stops the computation early when exceeding this budget and reports the classifier as skipped.	Float	0.5

[CLASSIFIER.WEAK_GOLDBACH]

Name	Function	Type	Default
MAX_LIST	Maximum number of triples $p+q+r$ to display.	Int	5
SHOW_ALL	Whether to list all ternary representations $n = p + q + r$ found within bounds.	Bool	false
SHOW_COUNT	Whether to include the total number of triples in the output.	Bool	true

[CLASSIFIER.ZECKENDORF]

Name	Function	Type	Default
SPARSE_MAX_TERMS	Maximum number of Fibonacci terms allowed in a <i>sparse Zeckendorf representation</i> . If the Zeckendorf decomposition of a number uses at most this many terms, it is classified as <i>sparse</i> . A value of 0 disables the sparse check.	Int	3

[CLASSIFIER.ZNAM]

Name	Function	Type	Default
MAX_BRANCH_PER_LEVEL	Maximum number of candidate children (branching factor) per search level. Keeps the backtracker from exploding.	Int	800
MAX_EXAMPLES	Maximum number of Znám sets to keep (only the first is normally printed).	Int	1
MAX_PRODUCT_BITS	Maximum allowed bit-length of the product $\prod d_i$ during search. Acts as a growth guard for candidate chains.	Int	512
MAX_WINDOW	Maximum window size for candidate generation when only one element is left to choose (final level).	Int	200
ODD_ONLY	Restrict the search to odd denominators only.	Bool	False
PREFIX_WINDOW	Base window size for candidate generation at intermediate levels (how far above the last element to scan for the next candidate).	Int	200
REQUIRE_PROPER	Require the chain to be a <i>proper</i> Znám set (i.e. $\sum 1/d_i = 1 - 1/\prod d_i$). If true, small k are rejected (k must be 5–10).	Bool	true
START	Starting integer for the Znám set search (seed element of the chain).	Int	2
TIME_BUDGET_S	Maximum time (in seconds) that any single classifier is allowed to spend on its computation. NumClass stops the computation early when exceeding this budget and reports the classifier as skipped.	Float	0.5

CLASSIFIER.ZUMKELLER]

Name	Function	Type	Default
BITSET_TARGET_CAP	Zumkeller partitioning requires finding a subset of divisors that sums to half of the total divisor sum, often written as $\sigma(n)/2$. BITSET_TARGET_CAP is the maximum allowed value of this half-sum for which the classifier will attempt the bitset dynamic-programming method. If $\sigma(n)/2$ is greater than this cap, the bitset would be too large, and the classifier abandons this method and returns skipped.	Int	10000000
DIVISOR_LIMIT	The maximum number of divisors for which NumClass will attempt any Zumkeller computation. If the number of divisors is greater than DIVISOR_LIMIT, the classifier immediately skips the number, because the divisor set is too large for safe processing.	Int	5000

Name	Function	Type	Default
SMALL_TAU_MITM	If the number of divisors is less than or equal to this value, NumClass uses the meet-in-the-middle (MITM) subset-sum solver, which is exact and very fast for small sets.	Int	32
WITNESS_TARGET_CAP	If $\sigma(n)/2$ is less than or equal to this value, and a valid partition is found, the classifier will also attempt to construct and display an explicit witness subset whose elements sum to $\sigma(n)/2$. If $\sigma(n)/2$ is larger than this cap, the classifier will still detect that the number is Zumkeller but will not attempt to produce the actual subset (to avoid heavy work).	Int	200000

Appendix C – Glossary of Mathematical Terms and Symbols

This glossary provides short, clear explanations of mathematical terms and symbols used throughout the NumClass manual. It is intended for readers who wish to understand key concepts in number theory and how they relate to the classifications.

Term	Definition / Explanation
Abundancy ratio	$\sigma(n) / n$, a measure of how “divisor-rich” a number is.
Additive persistence	Number of iterations needed to reach a single digit by repeatedly summing digits.
Aliquot cycle	A repeating loop in an aliquot sequence.
Aliquot sequence	Sequence $n, s(n), s(s(n)), \dots$ using the aliquot sum.
Aliquot sum $s(n)$	Sum of all proper divisors (that is, all positive divisors except n itself): $s(n) = \sigma(n) - n$.
Base-10 expansion / repeating decimal	Decimal representation of $1/n$; period = repeating block, preperiod = the non-repeating part.
Binomial coefficient	Number of ways to choose k items from n without order, written as $(n \text{ choose } k)$, often denoted $\binom{n}{k}$. $C(n,k) = \binom{n}{k} = n! / (k!(n-k)!)$
Carmichael function $\lambda(n)$	Smallest m with $a^m \equiv 1 \pmod{n}$ for all a coprime to n .
Centered figurate number	Figurate number with a central point and polygonal layers.
Clique	Set of graph vertices all mutually connected.
Combination	Selection of k items from n where order does not matter.
Composite number	Integer > 1 that is not prime.
Concatenation (:) 	NumClass operator joining integers: $12 : 34 \rightarrow 1234$.
Congruence	$a \equiv b \pmod{n}$ means n divides $(a - b)$.
Coprime (relatively prime)	$\gcd(a, b) = 1$.
Cycle (iterative)	Values that repeat during iteration of a function.
Cycle structure	Decomposition of a permutation into disjoint cycles.
Digital root	Repeated sum of digits until a single digit remains ($n \bmod 9$, with 9 instead of 0).
Digit reversal	Reversing decimal digits of n .
Divisor	d is a divisor of n if d divides n exactly.
Divisor count $\tau(n)$	Number of positive divisors of n .
Divisor sum $\sigma(n)$	Sum of all positive divisors of n .
Dynamic programming	Solving problems by storing intermediate results; used in bitset DP and combinatorial classifiers.
Euclidean algorithm	Method for computing $\gcd(a, b)$.
Even / Odd	Even $\Leftrightarrow$ divisible by 2; odd $\Leftrightarrow$ not divisible by 2.
Factorial $n!$	Product $n \times (n-1) \times \dots \times 1$.
Fibonacci sequence	$F_0=0, F_1=1, F_n=F_{n-1}+F_{n-2}$; appears in many areas of math.
Fixed point	Value x satisfying $f(x) = x$.
Floor $\lfloor x \rfloor$ / Ceiling $\lceil x \rceil$	Greatest integer $\leq x$ / smallest integer $\geq x$.
GCD (greatest common divisor)	Largest integer dividing both a and b .
Graph coloring	Assigning colors to vertices so adjacent ones differ.
Independent set	Graph vertices with no edges between them.
Iterative process	Repeatedly applying a function (e.g., Collatz, Kaprekar).

Jacobi symbol ($a n$)	Generalized quadratic-residue symbol. For odd n , $(a n)$ behaves like the Legendre symbol when n is prime and is computed from the prime factors of n . Used in primality and pseudoprime tests.
Lattice path	Path in a grid using permitted step types (used in Catalan/Motzkin).
LCM	Least common multiple of two integers.
Meet-in-the-middle (MITM)	Algorithm splitting a problem into halves for faster search (used in Zumkeller).
Modulo (mod)	Arithmetic where numbers “wrap around” a modulus.
Modular inverse	a^{-1} such that $a \cdot a^{-1} \equiv 1 \pmod{m}$ when $\gcd(a, m) = 1$.
Monochromatic	All items share the same color in a coloring.
Multiplicative function	$f(ab) = f(a)f(b)$ whenever $\gcd(a, b) = 1$.
Multiplicative persistence	Steps needed to reach a single digit via repeated digit multiplication.
Multiplicative order	Smallest k with $a^k \equiv 1 \pmod{m}$.
OEIS	Online Encyclopedia of Integer Sequences (https://oeis.org).
Ordered list / ordered subset	Selection of elements where order matters (used in Lah numbers).
Parity	Even/odd status of an integer.
Partition (integer)	Ways to write n as sum of positive integers ignoring order.
Partition (set)	Division of a set into disjoint non-empty subsets.
Peak value	Maximum value reached in a finite sequence.
Period / Preperiod	Repeating + non-repeating lengths of a decimal expansion.
Permutation	Arrangement of n elements in order.
Polygonal number	Number forming geometric regular polygons.
Prime factor count $\Omega(n)$	Total prime factors, with multiplicity.
Prime factor count $\omega(n)$	Number of distinct prime factors.
Prime factorization	Expression of n as product of primes.
Primitive root	g whose powers generate all nonzero residues mod p .
Proper divisors	Divisors of n excluding n itself.
Pseudoprime	Composite number passing certain prime tests.
Quasi-aliquot sequence	Iteration using $q(n) = s(n) - 1$.
Radical $\text{rad}(n)$	Product of distinct prime factors.
Reciprocal	$1/n$.
Recurrence relation	Sequence defined via previous terms.
Reduced aliquot sum $q(n)$	$q(n) = s(n) - 1$.
Reduced unitary divisor sum $q^*(n)$	$q^*(n) = s^*(n) - 1$.
Repdigit	Number consisting of repeated digits.
Repunit	Repdigit using only digit 1.
Residue class	Set of all integers $\equiv a \pmod{m}$.
Rough number	Integer whose <i>smallest</i> prime factor exceeds a bound.
Sequence	Ordered list of numbers defined by rule/iteration.
Set partition	See Partition (of a set).
Smooth number	Integer whose <i>largest</i> prime factor $\leq$ bound.
Smoothness parameter u	$\log n / \log P^+(n)$.
Squarefree	Integer not divisible by any square > 1 .
Totient $\phi(n)$	Number of integers $\leq n$ coprime to n .
Totient iteration	Repeated application of $\phi(n)$ until 1.
Unit fraction	Fraction of the form $1/n$.
Unitary divisor	A divisor d of n such that $\gcd(d, n/d) = 1$.
Unitary divisor count $\tau^*(n)$	Number of unitary divisors.

Unitary divisor sum $\sigma^*(n)$	Sum of all unitary divisors.
Weak order (total preorder)	Total, transitive relation allowing ties; used in Fubini numbers.
Zeckendorf decomposition	Unique representation of a non-negative integer as a sum of non-consecutive Fibonacci numbers.
Zero divisor (mod m)	$a \neq 0$ such that $a \cdot b \equiv 0 \pmod{m}$ for some $b \neq 0$.